

A Latch-On-Resistant, False-Alarm-Gated Lead-Time Metric for Bearing Anomaly Detection — and What It Reveals About SCADA-Rate Logging

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ABSTRACT

Bearing anomaly detectors are usually scored by classification accuracy, which ignores how early a warning arrives and at what false-alarm cost. We make lead time the primary evaluation object with a latch-on-resistant, false-alarm-gated metric: warning time counts only when the pre-onset false-alarm rate is within a budget τ , so a constant-on detector is provably invalid. With it we test whether a SCADA historian’s bin-averaged logging destroys lead time relative to decimation. Eleven detectors—four statistical process control (SPC) charts, Isolation Forest, three deep reconstruction autoencoders, a root-mean-square (RMS) trend baseline and two one-class models—run under a controlled sampling sweep on four inferential datasets: XJTU-SY ($n = 10$), FEMTO/PRONOSTIA ($n = 6$), University of Ferrara ($n = 6$) and NASA IMS ($n = 3$). On IMS the variance-sensitive charts shift positively ($3\sigma + 15.1$ h) but inconsistently across runs; no test survives Holm correction ($N = 44$; 0 of 44). Against a pre-specified ± 1 h margin, the resolution at which a planner can act, the difference is equivalent to zero in all 33 cells of the three multi-bearing campaigns, every 95% interval lying inside ± 0.6 h; IMS is too wide to decide in 9 of 11 cells, favours aggregation beyond the margin for CUSUM, and touches it for Hotelling T^2 (lower bound exactly $+1.00$ h). A single gas-turbine record of the Oil and Natural Gas Corporation (ONGC) cannot be tested. The campaigns that supply this statistical power are short-lived bearings, on which an IMS-sized effect could not appear, so the null bounds the effect at minutes, not hours. On these bearings deep models do not overtake the charts at any training fraction in $[0.20, 0.60]$; long-runway assets remain open. Code and results: <https://github.com/Ansh-Goyal01/Scada-Leadt ime-Benchmark>.

Keywords: bearing anomaly detection, lead-time metric,

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false-alarm gating, SCADA historian, prognostic horizon, statistical process control, conformal prediction

1. INTRODUCTION

Condition-based maintenance (CBM) and prognostics and health management (PHM) replace fixed-interval servicing with data-driven intervention, reducing both unplanned downtime and unnecessary maintenance (Jardine, Lin, & Banjevic, 2006; Lee et al., 2014; Lei et al., 2018). Rolling-element bearings are the most common failure mode in rotating machinery, and statistical control charts, classical machine learning and deep models are all used to detect incipient bearing faults from vibration (Randall & Antoni, 2011; Zhao et al., 2019). A warning delivered hours or days before a catastrophic seizure lets an operator schedule an orderly shutdown, order parts and avoid secondary damage to shafts, housings and couplings.

Yet most published detectors are evaluated with classification metrics—accuracy, precision/recall and receiver operating characteristic area under the curve (ROC–AUC)—which answer “was the failure detected?” but not the planner’s question: “how many hours of warning did we get, and how often did the system cry wolf?” A detector that warns 48 h ahead at a low nuisance-alarm rate is operationally distinct from one that fires minutes before breakdown or one that alarms every day. Time-aware prognostic metrics have long been advocated (Saxena et al., 2008; Saxena, Celaya, Saha, Saha, & Goebel, 2010; Goebel, Saha, & Saxena, 2011), yet lead time is rarely the headline quantity in detector benchmarks, and ad hoc definitions are easy to game: without a false-alarm constraint, a detector that alarms from the first sample—a “latch-on” exploit—earns the full run length as lead time while being operationally useless.

A second question is practical. Industrial Supervisory Control and Data Acquisition (SCADA) historians rarely store raw high-rate vibration; they keep decimated or time-averaged summaries at a logging interval of seconds

to minutes (Tautz-Weinert & Watson, 2017), not the tens of kilohertz at which accelerometers sample. The folklore concern is that bin-averaging these summaries washes out the excursions that signal an incipient spall, so that long-term storage would blind any prognostic algorithm that consumes the historian. To our knowledge this has not been tested with a sound lead-time metric across multiple datasets. If averaging is harmless, operators can store compact, cheap, bin-averaged streams without sacrificing warning time; if it is harmful, prognostics-grade monitoring requires separate high-rate acquisition.

We answer both questions. Lead time gated by a leakage-free pre-onset false-alarm rate is the primary metric, so the latch-on detector is provably scored invalid rather than winning. Under it we run a controlled sampling sweep that holds window content and alarm persistence constant, over nine headline detectors (four SPC charts, Isolation Forest, three deep reconstruction autoencoders and an RMS-trend baseline) plus two additionally evaluated one-class models, Deep SVDD (deep support vector data description) and a one-class support vector machine (SVM), on four inferential run-to-failure datasets—XJTU-SY ($n = 10$), FEMTO/PRONOSTIA ($n = 6$), University of Ferrara ($n = 6$) and NASA IMS ($n = 3$)—and a real single-asset ONGC gas-turbine SCADA record analyzed as a case study (Appendix D.2), with run-level significance tests (the run, not the within-run sampling factor, is the unit of inference) and Holm correction. The folklore is not supported on any dataset tested: historian-style aggregation does not destroy lead time. The difference is negligible on the multi-bearing sets, positive but inconsistent in direction on IMS, and consistent with non-destruction on the turbine record. The defensible conclusion is bounded non-destruction: no dataset $\times$ detector cell shows decimation superior beyond a ± 1 h operational margin, the resolution at which a planner can act (Section 6.4).

Contributions.

1. *A latch-on-resistant, false-alarm-gated, onset-relative lead-time metric*, crediting warning time only when the false-alarm rate over a data-anchored, leakage-free pre-onset region is within a stated budget. A constant-on detector is provably invalid and one firing just after the true onset is valid at zero false-alarm rate, by construction and by a released unit test. We do not claim the metric is unconditionally ungameable (Section 8).
2. *A formal comparison showing that gating changes the answer*, to our knowledge the first: the ungated Saxena prognostic horizon (PH) makes a different and operationally misleading recommendation (Section 6.11). On IMS the seven detectors PH ranks highest—the deep models, the one-class SVM, Hotelling T^2 , Isolation Forest and the exponentially weighted moving average (EWMA) chart—have no valid operating point at a 10% budget; the gated metric picks 3σ , PH's eighth.
3. *A controlled SCADA-sampling methodology* holding window content and alarm persistence constant, so an aggregate-vs-decimate difference reflects information loss rather than a window-counting artifact of coarser time grids.
4. *A multi-dataset null result*: with bootstrap confidence intervals and run-level tests, SCADA-rate aggregation does not destroy lead time on any of the four inferential datasets, and nothing survives Holm correction across the $N = 44$ family.
5. *Validation* of a conformal detector for distribution-free false-alarm control; a lead-time-vs-false-alarm trade-off analysis; feature-group and spectral ablations that localize the prognostic signal to time-domain statistics on IMS; and deep reconstruction models (a long short-term memory (LSTM) autoencoder, a temporal convolutional network autoencoder (TCN-AE) and a transformer anomaly detector) that do not become competitive with SPC charts at training fractions in $[0.20, 0.60]$ on short-lived bearings, a crossover on long-runway assets remaining open (Section 6.12).

Scope of claims. The results are for rolling-element bearings under approximately steady operating conditions; we make no claim of transfer to gearboxes, blades or variable-speed machinery, and the sampling question is specifically bin-averaged versus decimated historian storage. The metric itself is failure-mode-agnostic: it applies wherever a run-to-failure trajectory and a stationary normal region exist. Our working hypothesis was that bin-averaging the stored summary series would wash out incipient-fault excursions and shorten lead time. It does not, and we report the refutation, with its dataset-dependent direction explicit.

2. RELATED WORK

2.1. Prognostics, CBM, and Lead-Time Metrics

Reviews of machinery diagnostics and prognostics (Jardine et al., 2006; Lee et al., 2014; Lei et al., 2018) and of remaining-useful-life (RUL) estimation (Si, Wang, Hu, & Zhou, 2011) establish the CBM/PHM framing in which a health indicator is tracked and an alarm or RUL estimate is produced. Saxena et al. (2008), Saxena et al. (2010) and Goebel et al. (2011) propose prognostic-performance metrics (prognostic horizon, α - λ accuracy, relative accuracy and timeliness) that reward early, well-calibrated predictions. Our onset-relative lead time is in this spirit: it measures warning time relative to a degradation event, not an arbitrary positional marker. We depart from prior practice by making the false-alarm budget part of the metric's validity condition rather than a separately reported number, which closes the latch-on loophole, and we give what is to our knowledge the first formal comparison showing that an ungated prognostic horizon ranks

detectors differently, and misleadingly, from a false-alarm-budgeted metric (Section 6.11).

The cost asymmetry between a missed detection and a false alarm has been treated explicitly outside bearing prognostics. The Numenta Anomaly Benchmark (Lavin & Ahmad, 2015) scores streaming detectors under an application profile that rewards early detection and penalises false positives, making our gate’s commitment in another domain: earliness cannot be assessed independently of nuisance-alarm behaviour. The difference is structural. That benchmark folds both into one weighted score, so a detector may trade false alarms against earliness and still rank well; our budget is a validity condition, and an alarm outside it earns no credit that a longer lead can offset.

A parallel line of work revises evaluation itself for time-series detection. Tatbul, Lee, Zdonik, Alam, and Gottschlich (2018) generalise precision and recall from points to ranges with a tunable positional bias, so early detection within an anomalous interval can be rewarded, and Wu and Keogh (2023) argue that widely used benchmarks admit trivial solutions and overstate progress. Our metric shares the diagnosis but reports a physical quantity—hours of warning—that a planner can act on, rather than a score on a bounded scale, and the false-alarm constraint gates validity rather than weighting the score.

2.2. Anomaly Detectors and Control Charts

Univariate and multivariate statistical-process-control charts—the 3σ rule, the EWMA chart (Roberts, 1959) and Hotelling’s T^2 (Hotelling, 1947)—remain workhorses of industrial monitoring (Montgomery, 2009) because they are interpretable and need little training data. Isolation Forest (Liu, Ting, & Zhou, 2008) and the one-class SVM (Schölkopf, Platt, Shawe-Taylor, Smola, & Williamson, 2001) are widely deployed unsupervised learners; LSTM autoencoders (Hochreiter & Schmidhuber, 1997; Malhotra et al., 2016; Hundman, Constantinou, Laporte, Colwell, & Soderstrom, 2018) and Deep SVDD (Ruff et al., 2018) have been applied to machine-health monitoring (Zhao et al., 2019). We evaluate both families under one fixed hyperparameter protocol, so the comparison is about evaluation and sampling, not tuning.

Relation to published FEMTO/XJTU results. The IEEE PHM 2012 challenge and later RUL studies on FEMTO/PRONOSTIA and XJTU-SY (Wang, Lei, Li, & Li, 2020; Nectoux et al., 2012) report a RUL-error score on deliberately truncated Test and Validation copies. It is not comparable to ours: they estimate a continuous RUL on runs with no observed failure and report no false-alarm rate, so their scores say nothing about whether an alarm would survive our τ gate, whereas we measure early-warning lead time on complete run-to-failure trajectories; tabulating one

against the other would be a category error. The classic normalized-RMS health indicator with a first-prediction-time threshold, the baseline underlying most of that literature, is our `rms_trend` detector; re-evaluating specific published architectures under the gated metric (which requires their original code) is future work.

The pairing of detection delay with a false-alarm constraint is the founding formulation of sequential change detection. Lorden (1971) posed the minimax problem of minimising worst-case detection delay subject to a lower bound on the average run length to false alarm, Moustakides (1986) showed that the cumulative sum (CUSUM) chart of Page (1954) is exactly optimal under that criterion, and Pollak (1985) gives the corresponding result for a revised delay measure. Our gate inherits the premise but applies it when a detector is *scored*, against a rate measured on the data, rather than when it is *designed* by choosing a threshold that meets a run-length bound. The two can disagree: CUSUM has no valid operating point on IMS at a 10% budget, because its pre-onset false-alarm rate sits at 19.0–19.8% across every threshold we sweep.

2.3. Uncertainty Quantification and Conformal Prediction

Conformal prediction (Vovk, Gammerman, & Shafer, 2005; Laxhammar & Falkman, 2014) bounds the false-alarm rate by a target α under exchangeability, converting a raw score into an alarm with an interpretable, finite-sample guarantee. We wire conformal calibration into the lead-time pipeline and test the guarantee empirically across runs, including a run where exchangeability provably fails because the “normal” region already contains degradation.

2.4. Vibration Features

Bearing-defect frequencies, spectral kurtosis (Antoni, 2006) and envelope (demodulation) analysis (Randall & Antoni, 2011) are the classical frequency-domain tools for bearing diagnostics. We include fast Fourier transform (FFT) band energies, spectral kurtosis and Hilbert-envelope amplitudes at the bearing characteristic frequencies, and ablate them against time-domain statistics to see which family carries prognostic lead time under the gated metric.

2.5. SCADA-Based Condition Monitoring

Tautz-Weinert and Watson (2017) review SCADA-data condition monitoring and note that historian aggregation is a recognized but under-quantified concern: practitioners suspect that coarse averaging hurts fault detection, but the effect has not been measured with a principled warning-time metric. We address this gap and find that, for bearing faults, the concern is not borne out.

3. DATASETS

We use four inferential run-to-failure datasets and a single-asset real-historian case study (Table 1), chosen so the sampling claim is tested under different conditions: a high-rate raw-waveform rig where aggregation can be simulated faithfully (IMS), two multi-bearing campaigns (XJTU-SY and FEMTO/PRONOSTIA) that supply the statistical power a three-run rig cannot, a second modern constant-condition rig (Ferrara), and a historian stream that already is SCADA-rate data (ONGC, Appendix D.2). XJTU-SY ($n = 10$) is the primary inferential dataset; IMS ($n = 3$) is retained for its raw-waveform access, which permits the faithful aggregate/decimate simulation and the mechanism analysis (Appendix D.1).

IMS. The NASA/IMS dataset (Qiu, Lee, Lin, & Yu, 2006) provides raw 20.48 kHz accelerometer waveforms for three run-to-failure tests on a shaft loaded with four bearings, so frequency-domain features (spectral kurtosis, envelope) are meaningful; the coarsening sweep itself acts on the windowed feature vectors, not the waveform (Section 4.8). The failure label of IMS test 3 is set as documented in the released repository and justified in Appendix C.

XJTU-SY. The XJTU-SY dataset (Wang et al., 2020) contributes ten run-to-failure bearings under two operating conditions (five each) at 25.6 kHz, in 1.28 s snapshots once per minute. With $n = 10$ it gives the strongest generalization test here; its lifetimes range widely (52–533 min), stressing the metric with both long-degrading and abrupt-failure bearings.

FEMTO/PRONOSTIA. The FEMTO/PRONOSTIA platform (Nectoux et al., 2012) contributes six Learning_set bearings under three operating conditions, with horizontal and vertical accelerometers sampled at 25.6 kHz in 0.1 s snapshots every 10 s. Only the Learning_set bearings run to the platform’s 20 g vibration stop, so a failure, and hence a lead time, is observed; the Test_set/Validation_set copies are truncated mid-life for the IEEE PHM 2012 RUL challenge and would leave lead time undefined. Lifetimes span ~ 1.4 –7.8 h. We exclude the Paderborn dataset (Lessmeier, Kimotho, Zimmer, & Sextro, 2016): its pre-damaged bearings are measured at fixed conditions for fault classification, with no degradation timeline and no failure event, so it cannot test a lead-time or sampling claim, and using it would be a category error rather than a larger dataset count.

University of Ferrara. The Ferrara dataset (Arpa, Gabrielli, & Battarra, 2024) contributes an independent 2024 campaign: six self-aligning double-row ball bearings under constant speed and radial load, with continuous uniaxial vibration at 25.6 kHz over each bearing’s entire lifecycle (five-second contiguous snapshots). Lifetimes span 0.7–6.8 h, similar to FEMTO. Fixed operating conditions make it directly compatible with our onset detector without normalization;

Table 1. Datasets. “Role” is the function each plays in the argument: the four inferential datasets supply the run-level statistics, ONGC is a single-asset case study (Appendix D.2) and Paderborn is excluded.

Dataset	Role	Rate	Runs
IMS	controlled lab	20.48 kHz raw	3
ONGC	case study (App. D.2)	10 s historian	1
XJTU-SY	generalization	25.6 kHz, 1/min	10
FEMTO	generalization	25.6 kHz, 1/10 s	6
Ferrara	generalization	25.6 kHz, cont.	6
Paderborn	excluded (no RUL)	64 kHz	–

we use all six runs (E1–E6), duplicating the single channel into the two channel slots so the channel-invariant schema applies unchanged. Like FEMTO and XJTU-SY, its short lifetimes mean it cannot confirm or deny the IMS positive trend.

4. METHODOLOGY

4.1. Problem Formulation and Notation

For each run we observe snapshots indexed by time t . A feature extractor maps each snapshot (and a short trailing window) to $x_t \in \mathbb{R}^d$; a detector assigns a score s_t and, through a threshold and a persistence filter, an alarm indicator $a_t \in \{0, 1\}$. Each run has a failure time t_f and a train boundary t_b , and all scaling and threshold statistics are fit on $\{t \leq t_b\}$ only. We measure the lead time (how long before t_f the first sustained alarm fires) and the pre-onset false-alarm rate (how often the detector alarms while the machine is still healthy).

4.2. Channel-Invariant Feature Schema

Each snapshot yields per-channel time-domain statistics—RMS, kurtosis, skewness, peak-to-peak, crest factor and variance—and, for raw-waveform datasets, FFT band energies, spectral kurtosis (Antoni, 2006) and Hilbert-envelope amplitudes at the bearing characteristic frequencies (ball pass frequency outer race (BPFO), ball pass frequency inner race (BPFI), ball spin frequency (BSF) and fundamental train frequency (FTF)), computed from the dataset geometry.

A naive scheme that concatenates per-channel statistics and all cross-channel pairs is channel-count dependent and ill-posed: it gives 445 features for a 4-channel run and 1465 for an 8-channel run against only ~ 100 training snapshots ($p \gg n$), so it overfits, silently inflates the apparent dimensionality and is not comparable across runs. We instead aggregate each statistic family across physical channels (mean and max) and summarize each aggregated series over a rolling window by its mean, standard deviation, maximum and linear-trend slope, plus one cross-channel RMS-coherence feature: a fixed 49-dimensional space, identical for 4- and 8-channel assets and comparable across datasets (Appendix B). A train-fit variance top- k selection ($k \leq 50$) then guarantees $p < n$,

and all scaling is fit on the training split only, so no test-region statistic leaks into the features.

4.3. Leakage-Free Degradation Onset

Lead-time metrics that reference an arbitrary positional marker (e.g. “the first 5% of the test window is normal”) are gameable, because the marker is not tied to the physics of the run. We instead compute one degradation onset t_o per run from a scalar health indicator $h(t)$, by default the element-wise maximum of the baseline-standardized mean-RMS and mean-kurtosis trends, so it responds to whichever degradation mode (amplitude growth or impulsiveness) appears first:

$$h(t) = \max(z_b[\text{RMS}](t), z_b[\text{kurt}](t)), \quad (1)$$

where the baseline mean μ_b and standard deviation σ_b of the z-score $z_b[\cdot]$ are estimated only on data up to t_b , so t_o cannot see the test region. With band $\beta = \mu_b + k\sigma_b$ (default $k = 4$), the onset is the start of the last sustained excursion above β before failure,

$$t_o = \min\{t \in R_{\text{last}} : h(t) > \beta\}, \quad (2)$$

where R_{last} is the final run of consecutive above-band samples, merging gaps shorter than a tolerance and requiring a minimum persistence (Algorithm 1). Using the last sustained excursion makes the onset robust to early run-in transients and to the single trailing machine-off sample, either of which could otherwise mark a spuriously early onset.

Algorithm 1 Leakage-free terminal degradation onset

- 1: **Input:** health series $h(t)$; train boundary t_b ; failure t_f ; threshold multiplier k ; gap/persistence tolerances
 - 2: $\mu_b, \sigma_b \leftarrow \text{mean, std of } h \text{ on } \{t \leq t_b\}$
 - 3: $\beta \leftarrow \mu_b + k\sigma_b$
 - 4: $E \leftarrow \{t < t_f : h(t) > \beta\}$ ▷ above-band samples
 - 5: merge gaps in E shorter than tolerance; drop runs shorter than persistence
 - 6: $R_{\text{last}} \leftarrow \text{final surviving run in } E$
 - 7: **return** $t_o \leftarrow \min R_{\text{last}}$
-

The terminal-excursion rule has a predictable bias. For a degradation ramp of slope m the onset is late by approximately $k\sigma_b/m$, with a standard deviation of order σ_b/m . The persistence requirement P does not enter this bias: Algorithm 1 returns $\min R_{\text{last}}$, the *start* of the qualifying run, so P governs which excursions are admitted, not the timestamp reported for an admitted one. Bias and spread both scale inversely with the ramp slope: the estimator is near-unbiased and stable under abrupt degradation, where the delay falls to the sampling interval Δt , and systematically late and more variable under gradual degradation. This accounts for Table 3, where the abrupt first IMS test agrees across all three health indicators at 98–99% of span while the slow-degrading second and third tests move substantially.

The persistence (5 samples) and gap (10 samples) tolerances are fixed in samples of the health-indicator series, not physical time, and held constant across datasets, keeping the onset rule algorithmically identical everywhere at the price of physical-time comparability. At the native intervals (~ 10 min per snapshot for IMS after resampling to a uniform grid, 1 min for XJTU-SY, 10 s for FEMTO and ONGC) the 5/10-sample bars correspond to $\sim 50/100$ min on IMS, 5/10 min on XJTU and 50/100 s on FEMTO/ONGC. The difference is negligible for long records but not for the shortest bearings: XJTU Bearing2_4 (42 min total life) must clear a ~ 5 -minute persistence bar, roughly 12% of its lifetime.

Caveat: onset positions on bearings with total lifetime below 2 h are treated as fragile throughout; the per-bearing results of Table 7 report these bearings individually rather than discarding them.

The sensitivity analyses probe the threshold multiplier k (Table 3), not these timing tolerances, which stay in sample units for algorithmic reproducibility.

4.4. Onset-Relative Lead-Time Metric

Let the first sustained alarm time be FAT and the failure time t_f . The lead time is

$$L = \max\left(0, \frac{t_f - \text{FAT}}{3600}\right) \text{ hours}, \quad (3)$$

with $L = 0$ if there is no alarm or the alarm is at or after failure. The pre-onset false-alarm rate is the fraction of windows strictly before the onset that are flagged,

$$\text{FAR}_{\text{pre}} = \frac{|\{t < t_o : a(t) = 1\}|}{|\{t < t_o\}|}, \quad (4)$$

measured over the entire apparently-normal region rather than a positional sample. An alarm is valid if and only if

$$L > 0 \wedge \text{FAR}_{\text{pre}} \leq \tau, \quad (5)$$

with budget $\tau = 0.10$ throughout. If the pre-onset region is empty—the onset estimator places t_o at or before the first scored window— FAR_{pre} is undefined, the alarm cannot be gated, and it is excluded from the validity denominator rather than counted as valid, as is a run with no detectable onset.

Why the metric defeats the latch-on exploit. A detector that fires from the first sample earns the full run length as L but floods the pre-onset region, giving $\text{FAR}_{\text{pre}} \approx 1 > \tau$, so the conjunction in Eq. (5) scores it invalid; a detector that first fires just after t_o has $\text{FAR}_{\text{pre}} = 0$ and $L > 0$ and scores valid. A unit test in the released code asserts exactly these two cases: a constant-on detector is invalid, and an oracle firing at t_o^+ is valid at zero pre-onset FAR. Because t_o uses

only training-region statistics (Eq. 2), neither L nor FAR_{pre} can be tuned with test-region information. This defeats the specific latch-on exploit; it is not a claim of unconditional ungameability (Section 8).

4.5. Detectors

The headline set comprises nine detectors under one fixed hyperparameter protocol: an RMS-trend naive baseline, which alarms when the mean RMS exceeds $k\sigma$ above its training baseline and is included to verify that the metric penalizes a weak detector; the 3σ rule, alarming when the ℓ_2 -norm health index exceeds three training standard deviations; an EWMA chart (Roberts, 1959) ($\lambda = 0.2$, $k = 3.0$), sensitive to small sustained shifts; a deterministic two-sided tabular CUSUM chart (Page, 1954) ($k = 0.5$, $h = 5.0$) on the monitored magnitude, the classic change-point chart most sensitive to a small sustained mean shift such as slow degradation onset; Hotelling T^2 (Hotelling, 1947) in a principal component analysis (PCA)-reduced subspace ($\text{rank} \ll n$), so the covariance is well-conditioned; Isolation Forest (Liu et al., 2008), an unsupervised ensemble that isolates anomalies with few random splits; and three deep reconstruction baselines—an LSTM autoencoder (Malhotra et al., 2016), a temporal convolutional network autoencoder (dilated causal convolutions) and a lightweight reconstruction Transformer (Vaswani et al., 2017)—each trained on normal feature-window sequences only and scoring a window by its reconstruction error.

Additionally evaluated. Two one-class models complete the eleven: Deep SVDD (Ruff et al., 2018), whose multi-layer perceptron (MLP) encoder maps normal data toward a fixed center and scores a window by its squared distance to it, and the one-class SVM (Schölkopf et al., 2001). Deep SVDD is kept out of the headline set because it rarely alarms in time on short bearings (Section 6.12), but its numbers are reported in the main result tables because it is the low-false-alarm operating point under the gated metric (Section 6.9). The deep sequence models are data-starved: across FEMTO bearings and training fractions the runs furnish between 20 and 335 normal training windows (median 88 at the default split), and the largest IMS run 631. We include them as a modern-baseline comparison and to test whether a finite commissioning-data requirement makes them competitive with SPC charts (Section 6.12), without per-dataset tuning, the same fixed-protocol fairness applied to the charts. A run with fewer feature windows than the sequence length (short XJTU bearings, some coarse IMS factors) cannot form a sequence; that (run, detector) cell is recorded as an explicit N/A, not dropped. All non-conformal detectors alarm via a percentile threshold on training scores followed by a persistence filter, so alarm semantics are identical across detectors and only the score differs.

4.6. Conformal Calibration

A conformal detector (Vovk et al., 2005) wraps Isolation Forest: nonconformity scores are computed on a calibration split of normal data, and a test point alarms when its conformal p -value falls below α , which under exchangeability guarantees $\text{FAR} \leq \alpha$ in finite samples. We validate the guarantee empirically by measuring FAR_{pre} over the onset-defined normal region across a grid of α , and report the runs where it fails, with the reason.

4.7. SCADA Sampling Sweep

We coarsen the time grid by a factor $f \in \{1, 2, 5, 10, 20\}$ under two mechanisms: *aggregate* takes the mean within each bin of width f , exactly as a SCADA historian stores a bin-averaged value, and *decimate* keeps every f -th sample and discards the rest. A naive sweep changes the window's wall-clock span as f grows, which alters lead time through a window-counting artifact rather than information loss. Our controlled sweep holds window content and alarm persistence constant so that only the effective logging interval varies, isolating whether averaging loses fault-relevant information relative to decimation.

4.8. At What Level Coarsening Is Applied

The aggregate and decimate operations are different experiments depending on the level at which they act, so we state that level for each dataset. On no dataset is the raw waveform coarsened: every loader reduces each snapshot to per-channel time-domain statistics at ingest, so the sweep operates on summary series throughout. On IMS the sweep coarsens the windowed feature vectors, after rolling-window summarisation, which holds window content and alarm persistence constant as the effective logging interval varies. On XJTU-SY, FEMTO/PRONOSTIA, University of Ferrara and ONGC the sweep coarsens the per-snapshot feature series: *aggregate* takes the mean of the summary statistics within each bin, and *decimate* retains every f -th snapshot.

This is the operation a historian performs—historians store summary statistics, not waveforms—and it is the operation our claim concerns. It is not a test of raw-signal averaging. Averaging a raw waveform is a low-pass filter and does attenuate the impulsive transients that mark an incipient spall; we neither test nor dispute that, and Section 8 names it as the complementary study. The ONGC record is the clearest case, since its channels are the historian stream and no coarsening is simulated at all.

4.9. Statistical Protocol

Results are emitted in long format, one row per run $\times$ mode $\times$ factor $\times$ seed $\times$ method. Confidence intervals bootstrap runs, the unit of generalization; resampling seeds would understate

uncertainty.

The run is the unit of inference. A physical bearing run produces one degradation trajectory; the five coarsening factors applied to it ($f \in \{1, 2, 5, 10, 20\}$) are repeated transformations of that trajectory, i.e. pseudoreplicates. Pooling them as $3 \times 5 = 15$ “pairs” per detector, as an earlier version of this work did, inflates the effective sample size fivefold and yields overconfident p -values. We therefore collapse each run to one difference, the mean of (aggregate – decimate) over factors: $n = 3$ on IMS, $n = 10$ on XJTU-SY, and $n = 1$ on ONGC, a single asset for which we report descriptive differences only.

On these run-level differences we apply an exact two-sided sign test (binomial, $H_0 : P(\text{aggregate} > \text{decimate}) = \frac{1}{2}$), which uses only the direction of each difference and makes no normality or symmetry assumption that $n = 3$ could not support; a mixed-effects model with three random-effect levels would be over-fit and is deliberately not used. The test cannot reach significance at $n = 3$: all three runs concordant gives $p = 2(\frac{1}{2})^3 = 0.25$, so the IMS direction is a trend, not a significant effect. Runs whose collapsed difference is exactly zero are excluded from the sign-test count, so the n_+ and n_- tallies count non-zero-difference runs only. Across the detector $\times$ dataset family (ONGC excluded) we control the family-wise error rate with the Holm–Bonferroni step-down procedure at $\alpha = 0.05$, stating the family size N and the uncorrected FWER $= 1 - (1 - \alpha)^N$. One fixed hyperparameter protocol avoids optimistic per-condition tuning.

5. EXPERIMENTAL SETUP

All experiments use a fixed random-seed protocol; stochastic detectors (Isolation Forest, the conformal wrapper) inject the seed explicitly, so runs are bit-reproducible. Extracted features are cached to Parquet so that the sweep, ablations, calibration and trade-off analyses consume identical features. The environment is pinned to the package versions that produced these numbers, and a 181-test suite asserts the metric’s ungameability properties, the leakage-free onset and the conformal factory. Every figure and table is generated from the released result tables; no number is hand-edited.

6. RESULTS

IMS comes first for its raw-waveform access and the mechanism analysis it permits; XJTU-SY ($n = 10$) is the primary inferential test. Section 7 draws the results together.

6.1. Lead Time and Confidence Intervals on IMS

On raw (ungated) lead, with 95% bootstrap CIs over the $n = 3$ runs, the three deep reconstruction models top Table 2 with 183–198 h, followed by the one-class SVM, with Hotelling T^2 leading the control charts; the intervals are wide at $n = 3$, which is why the XJTU-SY, FEMTO and Ferrara campaigns

Table 2. IMS, all eleven detectors in prognostic-horizon order (PH rank 1–11). Raw: mean ungated lead (h) at full resolution and the default 97.5th-percentile threshold, with 95% bootstrap CI over the $n = 3$ runs. PH: best raw lead over the 95th, 99th and 99.5th percentiles. L_τ : best lead at an operating point whose pre-onset FAR, averaged across the three runs, is within τ (0.0 = none). † : no valid operating point at $\tau = 0.10$. Gated order at $\tau = 0.10$: 3σ , RMS-trend, Deep SVDD, then eight tied at 0. *: valid only at the loosest threshold and on one of three runs.

Detector	Raw	95% CI	PH	$L_{0.05}$	$L_{0.10}$	$L_{0.20}$
LSTM-AE †	197.6	[65.5, 331.7]	199.6	0.0	0.0	0.0
TCN-AE †	183.7	[65.5, 330.9]	185.4	0.0	0.0	0.0
Transformer-AD †	183.4	[65.5, 330.9]	184.0	0.0	0.0	0.0
One-class SVM †	180.2	[65.5, 316.7]	180.2	0.0	0.0	0.0
Hotelling T^2 †	174.8	[56.3, 315.9]	176.9	0.0	0.0	73.9
Iso. Forest †	87.2	[49.7, 152.2]	174.7	0.0	0.0	53.4
EWMA †	78.6	[28.0, 152.2]	89.1	0.0	0.0	0.0
3σ	64.6	[28.8, 108.7]	77.0	0.0	56.3	77.0
CUSUM †	64.8	[26.3, 114.9]	65.0	0.0	0.0	65.0
RMS-trend*	3.2	[0.0, 9.7]	34.7	34.7	34.7	34.7
Deep SVDD	14.8	[0.0, 34.0]	14.8	14.8	14.8	14.8

follow. The false-alarm budget overturns this ordering: the daggered detectors post their long leads only by flooding the pre-onset region and have no valid operating point under $\tau = 10\%$, leaving 3σ and Deep SVDD as the deployable choices (Section 6.11 makes the reordering formal). The deep models score $L = 0$ on all three IMS runs (valid-alarm fraction 0/3) because at the 97.5th-percentile threshold their pre-onset FAR ranges from 61% on test 3 to 100% on tests 1–2. Across the swept percentiles (Table 9) TCN-AE and Transformer-AD never attain a valid operating point, while LSTM-AE attains one at the 99.5th percentile on test 3 alone (59.67 h of lead at 4.19% pre-onset FAR, unchanged across ten random seeds), leaving its valid-alarm fraction at 1/3; its pre-onset FAR averaged across the three runs, 62.00%, is why the across-run gate of Table 2 scores it 0.0. Figure 1 shows the leakage-free onset and failure markers on a representative run, with the shaded training region from which the baseline statistics are computed.

6.2. Sensitivity of the Onset Definition

Onset stability on IMS. Across $k \in \{3, 4, 5\}$ the default onset moves by less than 0.1 percentage points of run span on test 1, 1.6 on test 3 and 5.4 on the slow-degrading test 2 (Table 3). On the first IMS test it sits at $\sim 98\%$ of span, only ~ 12.9 h before failure, so that run’s “pre-onset” region is almost the entire record and may already contain early degradation, which explains the conformal result below.

Short-bearing onset stability. Because the fixed-sample tolerances are a larger fraction of a sub-2-hour bearing than of a multi-week IMS run, we repeat the k sweep on all sixteen short XJTU and FEMTO bearings. Every one yields an onset at the default k , including the shortest (XJTU Bearing2_4

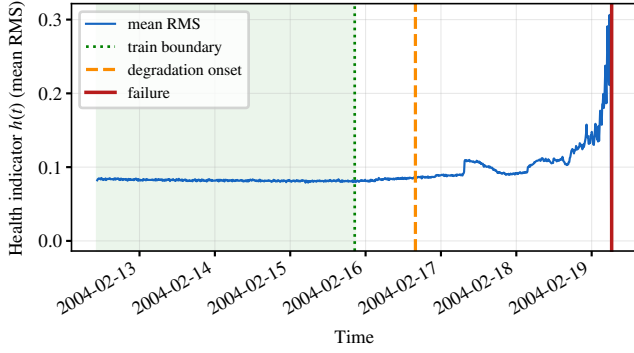


Figure 1. Health-indicator trajectory for IMS Bearing test 2 with the leakage-free degradation onset and failure markers. The baseline mean and standard deviation use only the shaded training region, so the onset cannot see the test data.

Table 3. Onset position on IMS, as % of run span, with the implied maximum lead (failure minus onset, h) in brackets. Left: the default RMS/kurtosis indicator with the terminal-excursion rule at threshold multiplier $k \in \{3, 4, 5\}$. Right: two decoupled indicators that do not share the amplitude basis of the magnitude detectors, at the default $k = 4$.

Run	rms_kurt (default)			Decoupled, $k = 4$	
	$k = 3$	$k = 4$	$k = 5$	kurt_only	pca1
Test 1	98.4 [12.9]	98.4 [12.9]	98.5 [12.7]	98.7 [10.4]	98.7 [10.7]
Test 2	60.4 [64.8]	61.9 [62.5]	65.8 [56.0]	98.8 [2.0]	83.6 [26.8]
Test 3	94.5 [59.3]	95.9 [44.5]	96.1 [42.2]	99.7 [3.5]	97.2 [30.0]

at 42 min, FEMTO Bearing3_1 at 1.43 h), and the median across- k spread is 1.0 percentage points of span on XJTU and 0.15 on FEMTO, versus 1.6 on IMS. The exceptions are XJTU Bearing1_2, which fires at only one of the three k values, and FEMTO Bearing1_1, with the widest spread (8.9 points); both are short-lived bearings the per-bearing analysis already flags as hard.

Onset under a decoupled health indicator. Because the default indicator shares its RMS/kurtosis basis with the detectors' features (Section 8), we recompute the onset under a kurtosis-only indicator and under the first principal component of the health channels (Table 3, right). The onset is not invariant: on the slow-degrading test 2 a kurtosis-only onset fires at 98.8% of span against the default's 61.9%, because amplitude rises before impulsiveness there, so absolute lead magnitudes depend on the indicator; test 1 agrees across all three definitions (~ 98 – 99% of span). The headline is unaffected, because the aggregate-vs-decimate contrast uses raw lead time (Section 4.9), which contains no onset term: re-running the full benchmark under the pca1 and kurt_only indicators leaves every raw-lead aggregate–decimate difference exactly unchanged, a maximum absolute deviation of 0.000 h across all 99 dataset $\times$ detector $\times$ indicator cells.

Alarm-persistence sensitivity. Over persistence $\in \{1, 3, 5, 10\}$ windows (default 3; Table 4a) the median

Table 4. Robustness on IMS and XJTU-SY. (a) Alarm persistence on IMS: run-level median aggregate–decimate lead-time difference (h) and the 3σ valid-alarm fraction, at full resolution and pooled over the five sampling factors, at persistence $\in \{1, 3, 5, 10\}$ windows, pooled over the 13 scoreable aggregate evaluations (Eq. 5). (b) Historian gaps: mean valid lead time (h) under random gaps of 0%, 5% and 20%, averaged over the runs with a valid alarm (Eq. 5) and then over five independent gap draws per level. XJTU-SY: the four bearings not scoreable at full resolution (Table 7) are excluded; $-$: no valid alarm on the other six.

(a) Alarm persistence (IMS)

Persistence (windows)	1	3	5	10
3σ median Δ (h)	+5.8	+15.1	+7.8	+3.8
EWMA median Δ (h)	+2.9	+2.1	+2.9	+2.3
Iso. Forest median Δ (h)	+0.3	+3.4	−0.8	+4.2
3σ valid-alarm frac., full res.	0.67	0.67	0.67	0.67
3σ valid-alarm frac., all factors	0.62	0.62	0.54	0.38

(b) Historian gaps

Detector	IMS lead (h)			XJTU-SY lead (h)		
	0%	5%	20%	0%	5%	20%
3σ	42.6	42.8	43.4	0.7	0.7	0.7
EWMA	41.8	41.8	41.7	1.6	1.6	1.3
CUSUM	39.7	39.3	39.0	1.4	1.5	1.3
Hot. T^2	186.1	186.1	134.9	—	—	—
Iso. Forest	54.7	54.3	53.8	0.7	0.7	0.7

direction is stable but its run-level consistency is not: 3σ has one negative run at the default ($n_+/n_- = 2/1$, $3/0$ elsewhere), EWMA is $2/1$ at persistence 1, 3 and 5 and $3/0$ only at 10, and Isolation Forest turns slightly negative at persistence 5. The 3σ valid-alarm fraction stays 0.67 at full resolution but, pooled over the sampling sweep, falls from 0.62 to 0.38 by persistence 10, so a longer filter costs the deployable chart valid alarms at SCADA rates.

Missing-data (historian-gap) robustness. Real historians drop records, so we set 5% and 20% of snapshot rows to missing (NaN, timestamp retained) and re-measure lead time on IMS and XJTU-SY over five independent draws per level (Table 4b), since a single random draw cannot separate a gap effect from the draw. At 20% every Table 4b detector with a valid alarm stays inside the ± 1 h margin on both datasets but one: the across-draw mean change is $+0.83$ h for 3σ on IMS and -0.02 h on XJTU-SY, -0.08 and -0.32 h for EWMA, -0.67 and -0.06 h for CUSUM, and -0.92 and -0.02 h for Isolation Forest; Hotelling T^2 has no valid alarm on a scoreable XJTU-SY bearing. The exception is Hotelling T^2 on IMS, which loses 51.2 h on average, bimodally across draws (0 to 128.1 h), according to whether a draw removes the rows carrying test 3's early alarm. It collapses in 2 of the 5 draws at 20%, giving an across-draw mean of 134.9 h over a range of 58.0–186.1 h, and the collapse is driven by a single run (test 3 falls from 315.9 to 59.7 h). At 5% missing, Hotelling T^2 and EWMA are unchanged in every draw on IMS, and no chart's per-draw mean lead moves by

more than 0.83 h. Valid-alarm fractions are unchanged on IMS (2/3 for every Table 4b detector at every gap level, in every draw) and move by at most two of the 6 scoreable bearings on XJTU-SY; apart from Hotelling T^2 on IMS at 20%, every cell of Table 4b varies across draws by less than 3.4 h on IMS and 0.87 h on XJTU-SY. We therefore claim gap-tolerance for the univariate charts and Isolation Forest—at most 1 h of lost warning at 20% missing rows on either dataset—and explicitly not for Hotelling T^2 , whose full covariance estimate can lose far more on a short-runway dataset, draw-dependently at $n = 3$.

6.3. Generalization to XJTU-SY ($n = 10$ bearings)

On the ten XJTU-SY bearings the IMS “aggregation helps” trend does not replicate (Table 5): the run-level median difference is 0 h for ten of the eleven detectors (-0.10 h for Isolation Forest), no sign test is significant ($p \geq 0.18$) and no direction is consistent across bearings, while per-run differences stay within a few minutes, so aggregation does not destroy lead time either. Absolute leads are short (0–3.5 h) because the bearings are short-lived (52–533 min): over the five window-magnitude detectors (3σ , EWMA, Hotelling T^2 , Isolation Forest and the RMS-trend floor, as in Table 7), 48 of 230 scoreable evaluations yield a valid alarm (220 of 450 are excluded under Eq. (5): 170 because the onset precedes the first scored window, 50 because Bearing1_2 has no detectable onset), and the shortest-lived and abrupt-failure bearings are hard or impossible to warn on under any mechanism. Only the three longest-lived bearings form a length-30 sequence, so each deep sequence model is N/A in 76/90 cells, a consequence of short runs rather than tuning; where they run, their difference is null too. Halving the sequence length to 15 windows makes only one more short bearing usable, Bearing2_5, whose onset precedes the first scored window, so it cannot be scored; on the two scoreable bearings the deep models still yield no valid alarm. Figure 2 summarizes the cross-dataset picture: a positive but non-significant median shift on IMS that is not consistent across its three runs, and null effects on XJTU-SY, FEMTO, Ferrara and ONGC, every interval inside the ± 1 h margin.

6.4. Generalization to FEMTO/PRONOSTIA ($n = 6$ bearings)

All six FEMTO Learning_set bearings are long enough for every detector, including the deep reconstruction models, to run on the full $n = 6$, and the verdict matches XJTU-SY (Table 5): every run-level median difference is within ± 9 min, only LSTM-AE and Transformer-AD are sign-consistent, both negative (0/3 and 0/4 non-zero runs), and the smallest sign-test p is 0.125 (Transformer-AD, raw), which does not survive Holm correction. Across the three multi-bearing campaigns all 33 dataset $\times$ detector cells are equivalent at $\delta = 1$ h, with no interval endpoint reaching ± 0.6 h, so the

Table 5. Run-level aggregate—decimate lead-time contrast on the three multi-bearing campaigns, collapsing the five sampling factors to one difference per run. Med., Mean: run-level median and mean difference (h), with the 95% bootstrap CI over runs; n_+/n_- : non-zero runs (s : all of one sign); p : exact two-sided sign test. ‡ : length-30 sequence models, $n = 3$ on XJTU-SY. Equivalent at $\delta = 1$ h when the CI lies inside $[-\delta, +\delta]$: all 33 cells. Holm over $N = 44$ (these 33 tests and the eleven IMS tests of Table 6): every adjusted p is 1.00. IMS and ONGC are outside the equivalence family.

Detector	Med.	Mean [95% CI]	n_+/n_-	p
<i>XJTU-SY</i> ($n = 10$)				
3σ	0.00	+0.030 [+0.000, +0.065]	3/1	0.625
EWMA	0.00	-0.002 [-0.012, +0.005]	2/2	1.000
CUSUM	0.00	-0.000 [-0.012, +0.013]	1/2	1.000
Hotelling T^2	0.00	+0.088 [+0.000, +0.260]	2/0 ^s	0.500
Iso. Forest	-0.10	-0.100 [-0.162, -0.041]	2/7	0.180
Deep SVDD	0.00	+0.015 [-0.038, +0.065]	3/2	1.000
One-class SVM	0.00	+0.006 [+0.000, +0.019]	1/0 ^s	1.000
LSTM-AE ‡	0.00	+0.056 [+0.000, +0.167]	1/0 ^s	1.000
TCN-AE ‡	0.00	+0.000 [+0.000, +0.000]	0/0	1.000
Transf.-AD ‡	0.00	+0.000 [+0.000, +0.000]	0/0	1.000
RMS-trend	0.00	-0.015 [-0.055, +0.015]	2/2	1.000
<i>FEMTO</i> ($n = 6$)				
3σ	-0.14	-0.213 [-0.449, -0.041]	1/5	0.219
EWMA	0.09	+0.078 [-0.005, +0.163]	4/2	0.688
CUSUM	0.10	+0.098 [-0.005, +0.207]	4/2	0.688
Hotelling T^2	0.07	-0.019 [-0.321, +0.194]	5/1	0.219
Iso. Forest	-0.07	-0.180 [-0.505, +0.025]	2/4	0.688
Deep SVDD	-0.00	-0.015 [-0.040, +0.002]	1/3	0.625
One-class SVM	0.02	-0.053 [-0.259, +0.075]	4/1	0.375
LSTM-AE	-0.04	-0.061 [-0.113, -0.013]	0/3 ^s	0.250
TCN-AE	0.00	-0.009 [-0.040, +0.014]	1/2	1.000
Transf.-AD	-0.05	-0.077 [-0.153, -0.018]	0/4 ^s	0.125
RMS-trend	-0.02	-0.123 [-0.327, -0.004]	1/3	0.625
<i>Ferrara</i> ($n = 6$)				
3σ	-0.01	-0.122 [-0.432, +0.080]	2/3	1.000
EWMA	0.01	-0.167 [-0.560, +0.046]	4/2	0.688
CUSUM	0.01	+0.024 [-0.001, +0.053]	5/1	0.219
Hotelling T^2	0.01	-0.078 [-0.351, +0.090]	4/1	0.375
Iso. Forest	0.00	-0.162 [-0.495, +0.049]	3/2	1.000
Deep SVDD	-0.00	-0.007 [-0.016, +0.000]	1/4	0.375
One-class SVM	0.01	-0.009 [-0.058, +0.023]	4/1	0.375
LSTM-AE	0.00	-0.073 [-0.255, +0.039]	2/2	1.000
TCN-AE	0.00	-0.089 [-0.292, +0.023]	3/1	0.625
Transf.-AD	0.00	-0.082 [-0.353, +0.091]	3/1	0.625
RMS-trend	0.01	+0.005 [-0.015, +0.023]	4/2	0.688

margin is cleared with room to spare. Exactly five cells have an interval lying wholly below zero—Isolation Forest on XJTU-SY, and 3σ (mean -0.21 h, $[-0.45, -0.04]$), LSTM-AE, RMS-trend and Transformer-AD on FEMTO—and every one favours decimation by less than the ± 1 h margin. These cells bound rather than overturn the conclusion: the direction of any residual effect varies by dataset, while the replicated conclusion is non-destruction. Across all eleven detectors 211/660 FEMTO evaluations yield a valid alarm; as on XJTU-SY, the short lifetimes (~ 1.4 – 7.8 h) keep absolute lead times small.

6.5. Generalization to University of Ferrara ($n = 6$ bearings)

The University of Ferrara campaign (Arpa et al., 2024), a second independent constant-condition multi-bearing rig, gives

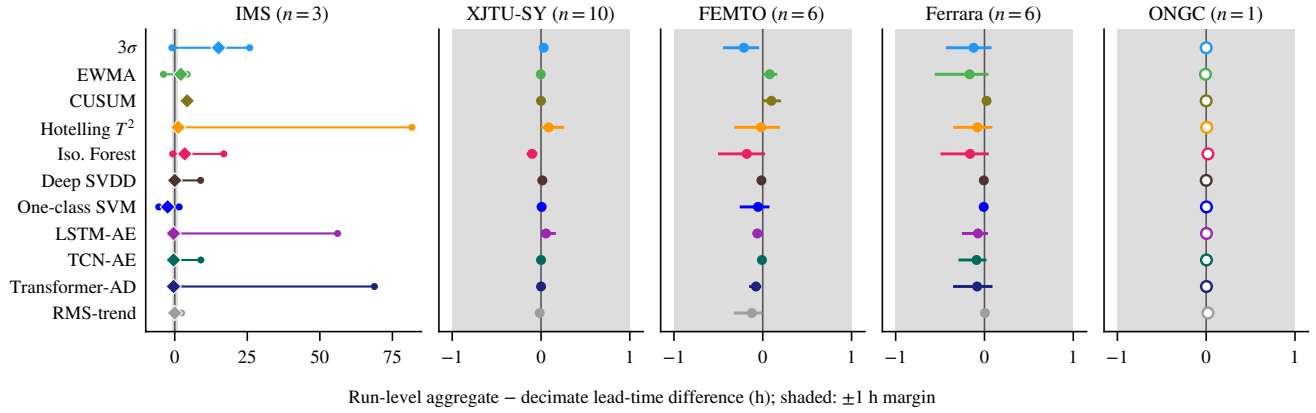


Figure 2. Run-level aggregate—decimate lead-time difference (h) per detector; the unit of inference is in each panel title. XJTU-SY, FEMTO, Ferrara: mean (dot) and 95% bootstrap CI over runs (Table 5). IMS, on a wider scale: the three per-run differences and their median (diamond). ONGC: the descriptive $n = 1$ median (hollow), a case study outside the four inferential datasets (Appendix D.2). IMS and ONGC values are in Table 6. Shaded: the ± 1 h equivalence margin.

the same verdict: every run-level median difference is within ± 1 min, no detector is sign-consistent across the six bearings, and the smallest sign-test p is 0.219 (CUSUM), not significant even before correction. Its 0.7–6.8 h lifetimes mean it cannot confirm or deny the IMS positive trend; it adds a second independent replication on a modern (2024) rig. The larger per-run negatives (-0.5 to -1.1 h) are all on E1, whose onset sits within ~ 0.1 h of failure (an abrupt, near-coincident-onset bearing like IMS test 1), not a systematic aggregation penalty.

6.6. Aggregate vs Decimate on IMS: a Positive but Inconsistent Median Shift

Under aggregation, IMS lead time stays high and stable as the logging interval coarsens; under decimation it degrades faster, with 3σ and Isolation Forest collapsing at coarse intervals (3σ falls from 64.6 h at full rate to 24.8 h at $10\times$ decimation while aggregation holds it near 64.6 h; Figure 3). The upturn at the largest factor is the few-windows regime, where only one or two runs survive and the mean is no longer comparable; we annotate rather than hide it.

At the run level (Table 6) the magnitude-monitoring detectors show a positive median difference, the largest for 3σ (+15.1 h), but only CUSUM and Hotelling T^2 are sign-consistent: aggregation shortens the lead time of 3σ , EWMA and Isolation Forest on one run each, and Deep SVDD is positive in one run and tied in two. The three deep reconstruction models and the one-class SVM have negative medians, each with two negative runs and one positive, the deep models' positive run being test 3; the RMS-trend baseline is mixed. With $n = 3$ the exact sign test cannot fall below $p = 0.25$, a floor only CUSUM and Hotelling T^2 reach, so nothing is significant or survives Holm correction: the defensible reading is a null result with a positive median direction for the magnitude-monitoring detectors. Against

Table 6. IMS controlled sweep ($n = 3$) and the ONGC case study ($n = 1$). IMS: run differences (h) for tests 1, 2, 3, each collapsing the five within-run sampling factors to one mean aggregate—decimate difference, with the run-level median, n_+/n_- and the exact two-sided sign-test p , which floors at 0.25 at $n = 3$ (*: one positive run and two ties, not counted as sign-consistent). ONGC: descriptive median over the five sampling factors, in minutes; no inference.

Detector	IMS (h)				ONGC	
	Run diffs	Med.	n_+/n_-	p	(min)	
3σ	+25.8, +15.1, -1.0	+15.1	2/1	1.00	0.0	
Iso. Forest	-0.7, +16.9, +3.4	+3.4	2/1	1.00	+1.2	
EWMA	-3.9, +4.3, +2.1	+2.1	2/1	1.00	-0.5	
CUSUM	+4.2, +4.7, +4.3	+4.3	3/0*	0.25	0.0	
Hotelling T^2	+1.0, +1.2, +81.7	+1.2	3/0*	0.25	+0.4	
Deep SVDD	+8.9, 0.0, 0.0	0.0	1/0*	1.00	0.0	
One-class SVM	-2.4, -5.5, +1.5	-2.4	1/2	1.00	+0.2	
RMS-trend	-0.2, 0.0, +2.4	0.0	1/1	1.00	+1.2	
LSTM-AE	-1.4, -0.4, +56.1	-0.4	1/2	1.00	+0.2	
TCN-AE	-1.4, -0.4, +9.0	-0.4	1/2	1.00	+0.2	
Transformer-AD	-1.4, -0.4, +68.8	-0.4	1/2	1.00	+0.2	

the ± 1 h margin the IMS intervals are too wide to decide for nine of the eleven detectors; CUSUM's lies wholly above it ($[+4.15, +4.67]$ h), favouring aggregation beyond the margin, and Hotelling T^2 's lower bound sits exactly on it ($[+1.00, +81.74]$ h). The candidate mechanism is that bin-averaging acts as a low-pass filter suppressing single-window noise in the health statistic, so the alarm threshold is crossed earlier and held more persistently than under decimation; Appendix D.1 tests it and Section 7.3 gives the signal-theoretic reason.

6.7. Multiple-Comparison Correction

The aggregate-vs-decimate question is asked once per detector on each inferential dataset, a family of $11 \times 4 = 44$ tests (ONGC excluded) whose uncorrected family-wise error rate at $\alpha = 0.05$ is $1 - (1 - 0.05)^{44} \approx 0.90$. Under Holm—

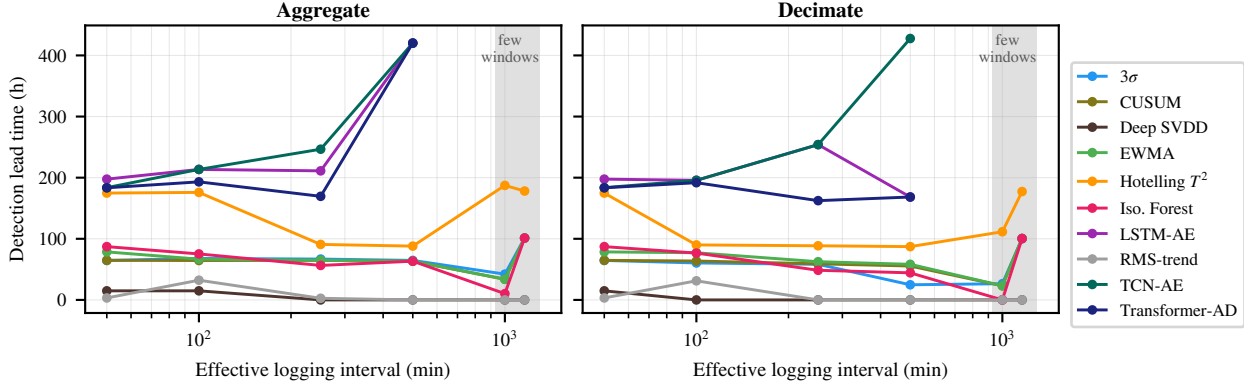


Figure 3. Detection lead time vs effective SCADA logging interval on IMS (controlled sweep, mean across runs): aggregation (left) and decimation (right). Shading marks the few-windows regime at the two largest factors, where only one or two runs remain.

Bonferroni no hypothesis survives (0 of 44): the smallest raw p is 0.125 (Transformer-AD on FEMTO), and every one of the forty-four adjusted p -values is 1.00. After honoring the run structure and correcting for multiplicity there is no significant aggregate-vs-decimate effect on any dataset; the positive IMS median shift remains a trend that cannot be significant at $n = 3$. The family is heterogeneous in power—the three deep sequence models contribute $n = 3$ XJTU tests resting on the longest-lived, most warnable bearings († in Table 5)—but every uncorrected p already exceeds 0.1, so this changes no rejection decision; we flag it so $N = 44$ is not read as forty-four equally powered tests.

Per-bearing heterogeneity. Table 7 breaks XJTU-SY down by bearing at full resolution: the mean lead over the valid detectors is the realistic single-deployed-detector expectation, and the oracle “best” presumes foreknowledge of which detector fires first; they differ only where valid detectors disagree, which happens on none of the five scoreable bearings (Table 7 marks the other five and why). The 48/230 valid-alarm figure hides strong heterogeneity. Long-degrading bearings such as Bearing2_3 (533 min life) and Bearing2_1 (491 min) are warnable, with 3.45 and 0.67 h of lead, whereas two abrupt-failure bearings, Bearing2_4 (42 min, the shortest-lived) and Bearing1_1 (123 min, sudden end), yield no valid alarm for any detector (2 of the 5 scoreable bearings): their onset and failure nearly coincide, no method can manufacture warning time the physics does not provide, and the metric credits none.

The same contrast under the gated metric. The headline uses raw lead, the quantity coarsening acts on and one independent of τ . Because we argue that ungated lead can mislead, we also run the contrast under the gated metric itself (Table 8). The answer does not change: under either metric every inferential dataset’s median run-level difference is at most a few thousandths of an hour, no detector is sign-consistent enough to reach significance, and no hypothesis survives Holm correction across the $N = 44$ family. What changes is power: gat-

Table 7. XJTU-SY per-bearing summary at full resolution (aggregate). V/5 counts the detectors (of five) that produce a valid alarm (Eq. 5); Mean is the mean lead over those detectors; Best is the oracle maximum over them. n.s.: not scoreable, and excluded from the totals—^ano detectable onset; ^bonset at or before the first scored window, so the pre-onset region is empty.

Bearing	Cd.	Life (min)	V/5	Mean (h)	Best (h)
Bearing1_1	1	123	0	—	0.00
Bearing1_2	1	161	n.s. ^a	—	—
Bearing1_3	1	158	n.s. ^b	—	—
Bearing1_4	1	122	2	0.68	0.68
Bearing1_5	1	52	n.s. ^b	—	—
Bearing2_1	2	491	3	0.67	0.67
Bearing2_2	2	161	n.s. ^b	—	—
Bearing2_3	2	533	1	3.45	3.45
Bearing2_4	2	42	0	—	0.00
Bearing2_5	2	339	n.s. ^b	—	—
Total (5 scoreable bearings)					6/25
Scoreable bearings with no valid detector					2/5

ing discards every run without a valid alarm, so the mean effective sample size per detector falls from 3.0 to 1.3 on XJTU-SY, 4.8 to 2.8 on FEMTO, 5.0 to 3.6 on Ferrara and 2.7 to 1.2 on IMS, roughly a halving. The gated contrast agrees with the raw one with materially less power, so raw lead stays the primary quantity and the gated version is a robustness check.

6.8. Conformal Calibration

On IMS (Figure 4a, values in its caption) the distribution-free guarantee $\text{FAR} \leq \alpha$ holds on test 2 (0% through $\alpha = 0.10$) and on test 3 through $\alpha = 0.05$ (0.9% at $\alpha = 0.01$, 4.4% at 0.05), slightly exceeding the bound at $\alpha = 0.10$, where the empirical FAR is 13%. It fails on test 1, at 0.74–0.85 regardless of α , for the reason Table 3 foretold: the onset fires only ~ 12 h before failure, so the “pre-onset” region already contains degradation and calibration and “normal” scores are not exchangeable. We keep this run because it delimits when

Table 8. Aggregate—decimate contrast under the raw and the gated ($\tau = 10\%$ pre-onset FAR) metric, all eleven detectors. Med.: median across detectors of each detector’s run-level median difference (h); $n_+/n_-/0$: detectors by the sign of that median; $\bar{n}_{\text{eff}}$: mean effective sample size once zero-difference runs are dropped.

Dataset	Raw lead			Gated lead		
	Med.	$n_+/n_-/0$	$\bar{n}_{\text{eff}}$	Med.	$n_+/n_-/0$	$\bar{n}_{\text{eff}}$
XJTU-SY	+0.000	0/1/10	3.0	+0.000	0/1/10	1.3
FEMTO	−0.004	4/6/1	4.8	+0.000	3/2/6	2.8
Ferrara	+0.005	8/2/1	5.0	+0.000	1/5/5	3.6
IMS	+0.000	5/4/2	2.7	+0.000	4/0/7	1.2

the guarantee is usable in prognostics: it needs a genuinely stationary normal region, which an abrupt late-onset failure does not provide. On XJTU-SY (Figure 4b) bearings with a substantial stationary pre-onset region (e.g. Bearing2_3, calibrated through $\alpha = 0.05$) honor the bound, while those with only one or two pre-onset windows, an artifact of short lifetimes, are flagged uncalibrated rather than silently passed. The conformal machinery thus doubles as a per-asset usability test of whether enough stationary normal data exists to control the false-alarm rate at all.

FEMTO and ONGC calibration. On FEMTO (Figure 4c) no bearing honors the bound at $\alpha = 0.05$ (pre-onset FAR 0.08–0.56; pooled 0.27): the short 1.4–7.8 h lifetimes leave a pre-onset region that is neither long nor stationary. The most stationary bearings come closest (Bearing2_2 at 0.08, Bearing1_2 at 0.10) and the abrupt Bearing1_1 (0.56) violates it badly. On the real ONGC historian stream (Figure 4d) the calibration tracks the diagonal closely (empirical FAR 0.016, 0.030 and 0.096 at $\alpha = 0.01, 0.02, 0.05$), a modest 1.5–2 $\times$ above target, consistent with a mostly-stationary record that only mildly violates exchangeability. The guarantee is therefore usable where a stationary normal region exists (ONGC, the longer IMS/XJTU bearings) and self-reports as invalid where it does not (short FEMTO/XJTU bearings, abrupt onsets).

6.9. Lead-Time-vs-False-Alarm Trade-off

Sweeping the alarm-threshold percentile traces a prognostics analog of the ROC curve (Figure 5, Table 9): tightening the threshold lowers pre-onset FAR and lead together, so an operator can select an operating point for a stated false-alarm budget. On the achievable frontier Hotelling T^2 and Isolation Forest dominate the upper left, while the naive RMS-trend baseline collapses to near-zero lead once the threshold tightens (1.3 h at the 99.5th percentile). Hotelling T^2 has the highest raw lead of any control chart but exceeds the $\tau = 10\%$ budget at every threshold tested, so it has no valid operating point on IMS; EWMA and CUSUM behave the same way, their pre-onset FAR sitting at 19.0–22.1% across the tabulated thresholds (Table 9), so neither memory-based chart should be recommended under a 10% budget. Deep SVDD occu-

pies the opposite corner, within budget ($\leq 0.6\%$ FAR) only by firing late (≤ 14.8 h of lead). 3σ reaches a valid point, 56.3 h of lead at 8.9% FAR at the 99.5th percentile, and is the recommended chart under a 10% budget, a ranking that the raw-lead column of Table 2, led by the deep models, would by itself obscure. The ten XJTU bearings (Figure 5b) give the same qualitative shape at shorter absolute leads, so the operating-point selection generalizes beyond the laboratory rig.

Sensitivity to the false-alarm budget. The budget τ is a policy input, and the recommendation is not τ -invariant (Table 2). At $\tau = 0.20$ the 3σ chart gives the longest valid lead (77.0 h), joined by CUSUM, Hotelling T^2 and Isolation Forest as their pre-onset FAR falls within the looser budget; at $\tau = 0.10$ it remains the best valid chart (56.3 h); at a strict $\tau = 0.05$ no control chart qualifies, and only Deep SVDD (14.8 h) and the RMS-trend floor (34.7 h) stay within budget, so an operator with very low nuisance-alarm tolerance should prefer the low-FAR Deep SVDD point. The 3σ recommendation therefore holds for $\tau \geq 0.10$ on IMS and inverts toward the low-FAR detectors below it. The cross-dataset headline, computed on raw lead-time differences (Section 4.9), does not depend on τ at all.

Where does $\tau = 10\%$ come from? Consider an expected-cost model with two site parameters: c_I , the cost of responding to one nuisance alarm (a wasted inspection), and c_F , the cost of an unplanned bearing failure (downtime plus secondary shaft/housing damage). Over a pre-onset life of k windows a detector at pre-onset FAR f issues $\approx kf$ nuisance inspections, and a valid alarm with adequate lead converts the failure into a planned repair, saving c_F . False alarms are worth tolerating up to the budget at which their cumulative cost equals the warning’s value, $k\tau c_I \approx c_F$, i.e.

$$\tau^* \approx \frac{c_F}{k c_I} = \frac{\rho}{k}, \quad \rho \equiv c_F / c_I. \quad (6)$$

For rolling-element bearings an unplanned failure typically costs $\rho = 10^2$ – 10^4 times a routine inspection, and the pre-onset window count is of order $k \sim 10^3$ on these runs, so Eq. (6) places the break-even budget at $\tau^* \sim 0.1$ – 1 (τ^* saturates at 1 for $\rho \geq 10^3$ and falls to 0.10 at $\rho = 10^2$, $k = 10^3$). Our $\tau = 0.10$ is the conservative end, justified whenever failures are at least $\sim 100\times$ costlier than inspections; the strict $\tau = 0.05$ column corresponds to inspections that are themselves disruptive (e.g. requiring a shutdown), a small ρ , exactly where the low-FAR Deep SVDD point becomes economical. This is an order-of-magnitude anchor, not a site-specific optimisation: the true optimum needs an operator’s actual c_I and c_F , which we do not have, and Eq. (6) lets a practitioner substitute their own.

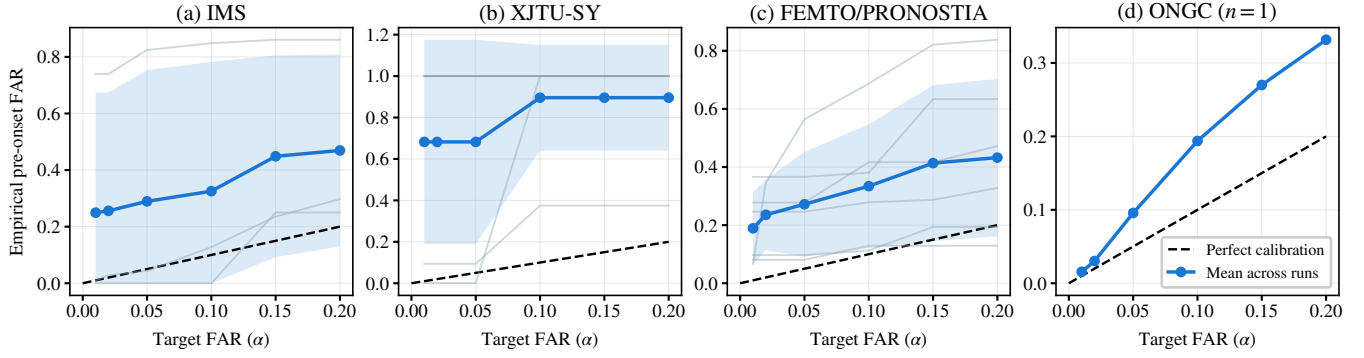


Figure 4. Conformal calibration: empirical pre-onset FAR vs target α for (a) IMS, (b) XJTU-SY, (c) FEMTO/PRONOSTIA and (d) the ONGC turbine ($n = 1$ case study, Appendix D.2). Points on or below the diagonal satisfy the distribution-free guarantee; bearings with only one or two pre-onset windows are flagged uncalibrated. Panel (a) empirical FAR at $\alpha = 0.01, 0.05, 0.10, 0.20$: test 1 0.74, 0.82, 0.85, 0.86; test 2 0.00, 0.00, 0.00, 0.25; test 3 0.01, 0.04, 0.13, 0.30.

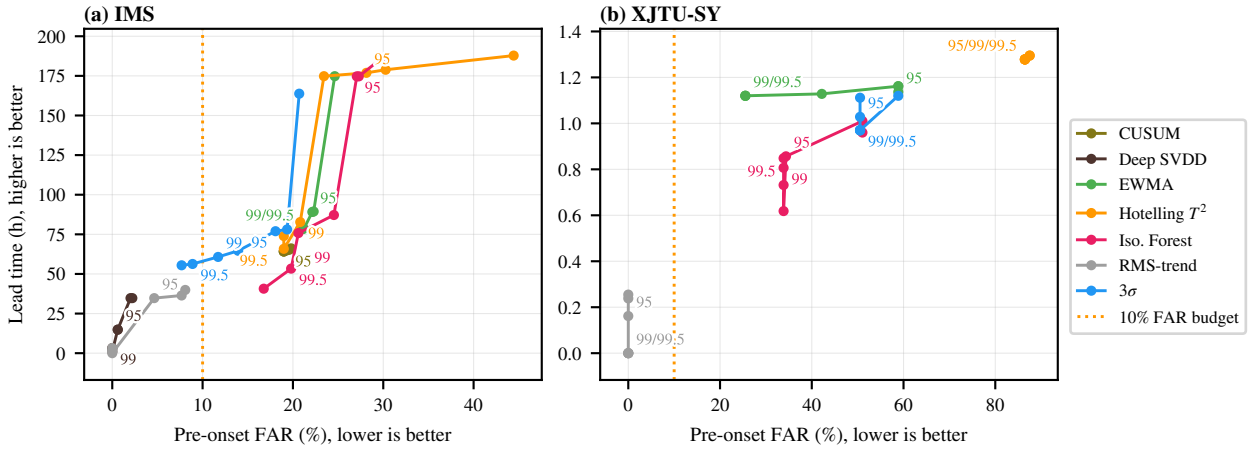


Figure 5. Lead-time vs pre-onset-FAR trade-off, threshold percentile swept (mean across runs): (a) IMS, (b) the ten XJTU-SY bearings; labels mark the 95th, 99th and 99.5th-percentile operating points (tabulated for IMS in Table 9) where they fit legibly. Up and to the left is better. The deep reconstruction models and the one-class SVM, above the budget at every threshold, appear in Table 9 only.

6.10. Feature-Group Ablation

On IMS, time-domain amplitude features (`rms_only` or `time_only`) give each of the six non-sequence detectors its highest or tied-highest validity, and adding spectral, envelope or all features never improves validity for any of them (Table 10). For the SPC charts the effect is sharp: `rms_only` is the unique group at which 3σ and EWMA reach full validity (1.00 at 0% pre-onset FAR), while spectral, envelope and all collapse all four charts to 0.33 by driving their pre-onset FAR to 42–63%, the variance-sensitive charts reacting violently to noisy extra dimensions. Deep SVDD fails the opposite way: its pre-onset FAR stays near 0% under every feature set, yet spectral features cut its validity to 0.33 (0.00 for `spectral_only`) by delaying or suppressing the alarm—missed detections, not false alarms. Under a false-alarm budget every detector should therefore be restricted to time-domain amplitude features, and frequency-domain features admitted

with care, most of all for the SPC charts. Raw lead inverts this ranking without overturning it: on Isolation Forest the envelope group gives the longest mean lead (99.4 h) and `rms_only` the shortest (32.0 h), but the long-lead groups accrue 23–29% pre-onset FAR against 0.2% for `rms_only`, buying their lead by firing indiscriminately; per run, spectral features flip 3σ , EWMA and Hotelling T^2 from valid to invalid on test 2 at 100% pre-onset FAR while Isolation Forest stays valid in both settings. Repeated at every coarsening factor, `rms_only` stays uniquely best at native and $2\times$ resolution ($f \leq 2$), matching the argument that bin-averaging preserves the amplitude statistic; at $f \geq 5$ only a handful of windows survive per run, valid fractions move in steps of $1/3$ ($1/2$ at $f = 20$, where test 2 is not scoreable), and whether the advantage holds at the coarsest SCADA rates is not resolvable at $n = 3$.

Table 9. IMS lead-time-vs-false-alarm trade-off (mean across runs) at three threshold percentiles: lead (Ld, h) and pre-onset FAR (%). †: pre-onset FAR above the $\tau = 10\%$ budget, so the operating point is invalid under the gated metric. The three deep reconstruction models and the one-class SVM are listed so the “no valid operating point” claim can be checked per threshold: all twelve of their entries are daggered.

Method	95th		99th		99.5th	
	Ld	FAR	Ld	FAR	Ld	FAR
3σ	77.0	18.1†	60.7	11.7†	56.3	8.9
EWMA	89.1	22.1†	78.3	21.0†	78.3	21.0†
CUSUM	65.0	19.4†	64.2	19.0†	64.2	19.0†
Hot. T^2	176.9	28.1†	82.7	20.8†	73.9	19.0†
Iso. For.	174.7	27.1†	75.8	20.6†	53.4	19.8†
Deep SVDD	14.8	0.6	3.2	0.0	2.7	0.0
RMS-trend	34.7	4.6	1.3	0.0	1.3	0.0
LSTM-AE	199.6	87.8†	194.0	82.2†	93.6	62.0†
TCN-AE	185.4	82.6†	183.1	79.9†	180.4	74.4†
Transf.-AD	184.0	81.1†	181.7	79.2†	178.7	72.0†
OC-SVM	180.2	63.2†	92.5	56.7†	92.5	55.4†

Table 10. Feature-group ablation on IMS: valid-alarm fraction (mean over the three runs) by feature group, for the six non-sequence detectors. EW: EWMA; CU: CUSUM; DS: Deep SVDD; IF: Isolation Forest.

Group	3σ	EW	CU	T^2	DS	IF
rms_only	1.00	1.00	0.67	0.67	0.67	1.00
kurt_only	0.67	0.67	0.33	0.67	0.33	0.67
time_only	0.67	0.67	0.67	0.67	0.33	0.67
spectral_only	0.33	0.33	0.33	0.33	0.00	0.67
envelope_only	0.33	0.33	0.33	0.33	0.33	0.67
time+spectral	0.33	0.33	0.33	0.33	0.33	0.67
all	0.33	0.33	0.33	0.33	0.33	0.67

6.11. Comparison with the Saxena Prognostic Horizon

If the false-alarm gate is what separates a usable warning from a manufactured one, a metric without it should reward detectors ours rejects. We test this against the most widely used ungated warning-time metric, the Saxena prognostic horizon (Saxena et al., 2008, 2010): $PH = \max(0, (t_f - FAT)/3600)$ hours, with no false-alarm constraint. It is exactly our lead time L before the validity gate (Eq. 5) is applied, and that one missing conjunct is decisive.

PH is maximized by the latch-on detector. On the three IMS runs the constant-on detector earns the full run length as PH (164–1073 h) while flooding the pre-onset region ($FAR_{pre} \approx 1$), so the gated metric scores it $L = 0$ on every run; the onset oracle is the mirror image, with zero pre-onset false alarms and the maximum achievable lead (12.9, 62.5 and 44.5 h).

PH and L disagree on deployable detectors too. Ranked at their own best operating points (Table 2), the two orders nearly invert at the top. The seven detectors PH ranks highest—LSTM-AE 199.6 h, TCN-AE 185.4 h,

Transformer-AD 184.0 h, the one-class SVM 180.2 h, Hotelling T^2 176.9 h, Isolation Forest 174.7 h and EWMA 89.1 h—have no valid operating point under a 10% false-alarm budget, so their gated lead is $L = 0$; the deep models alarm through 61–100% of every pre-onset region. 3σ , which PH ranks only eighth (77.0 h raw), is the one that clears the budget (56.3 h of lead at 8.9% pre-onset FAR) and the correct recommendation; CUSUM (65.0 h, PH’s ninth) also scores $L = 0$, and Deep SVDD, ranked last by PH, rises to a valid 14.8 h.

Implication. Any warning-time metric that does not gate on FAR_{pre} can be driven to its maximum by loosening the alarm threshold, in the limit to the constant-on detector; ours closes this by construction and PH does not. On IMS this changes the recommended detector from the ones PH ranks first by raw lead to 3σ , the only classical chart with a valid operating point at a 10% budget: PH rewards precisely the detectors a false-alarm-budgeted operator must avoid.

6.12. Minimum Training Data for Deep Anomaly Detectors

How much normal run-in history must a new asset collect before a deep model can be trusted over a simple control chart? We sweep the training fraction $T \in \{0.20, \dots, 0.60\}$ on FEMTO, the one dataset where all six bearings are long enough for every detector, including the seq-len-30 deep models (calibration fixed at 0.10, test = $1 - T - 0.10$, no per- T retuning). At $T = 0.20$ the shortest bearing (Bearing3_1) forms only 20 training windows, fewer than the 30 a sequence requires, so LSTM-AE, TCN-AE and Transformer-AD are explicit N/A there (Section 4.5); these are the only three N/A cells among the 300 (detector, bearing, T) cells. Valid-alarm fractions are over $n = 6$ bearings, or five at $T = 0.20$, where Bearing2_2’s onset precedes the first scored window (Eq. 5); an N/A counts as no valid alarm, the conservative choice; the released sweep records the N/A count separately.

There is no crossover (Figure 6). More commissioning data does not systematically improve the four SPC charts: 3σ is at its best at the *smallest* training fraction (1.00) and falls to 0.833 at $T = 0.30$ and 0.667 from $T = 0.40$ on; Hotelling T^2 starts at 0.600, falls to 0.333 at $T = 0.30$ and settles at 0.500; EWMA stays within 0.600–0.667; and CUSUM stays within 0.400–0.500 apart from a one-bearing rise to 0.667 at $T = 0.50$. Isolation Forest starts at 0.600, dips to 0.167 at $T = 0.30$ and ends at 0.667; only the naive RMS-trend baseline rises steadily with T , from a far lower base (0.000 to 0.500). This is a consequence of the design, not reverse data-hunger: the test fraction $1 - T - 0.10$ shrinks as T grows, so on bearings this short a better-estimated baseline comes with a shorter window in which to use it. The three deep reconstruction models sit far below the charts at every train-

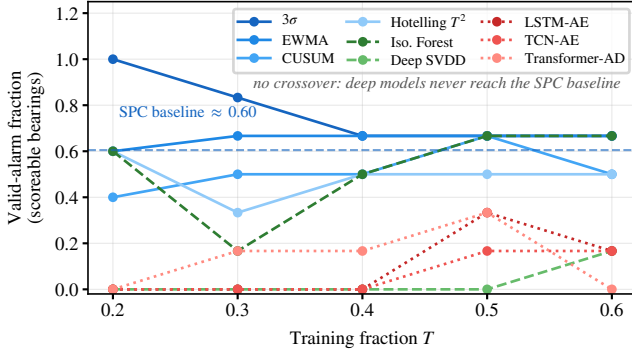


Figure 6. Valid-alarm fraction vs. training fraction T on FEMTO ($n = 6$): SPC charts in blue, deep reconstruction models in red, Deep SVDD separately; dashed: the SPC baseline (≈ 0.60). The three N/A cells at $T = 0.20$ (Bearing3_1, 20 training windows) count as no valid alarm; $T = 0.20$ is over five bearings (Section 6.12).

ing fraction (LSTM-AE and Transformer-AD 0.000–0.333, TCN-AE 0.000–0.167) and do not close the gap. Deep SVDD has no valid alarm below $T = 0.60$ and 0.167, one bearing, at $T = 0.60$, the shortest test window: as on IMS test 1 it fires very late, because its threshold requires substantial encoder-distance growth from the trained center, so on these 1.4–7.8 h bearings its first alarm arrives at or after failure on all six bearings below $T = 0.60$ and on five of the six at $T = 0.60$. That is a bearing-lifetime constraint, not a training-data constraint.

On bearings of the lifetimes tested here, control charts should therefore be preferred regardless of how much run-in history is available, because the deep reconstruction models do not overtake them anywhere in the $[0.20, 0.60]$ training range. This is a statement about a data regime, not the architectures: across bearings and training fractions these runs furnish between 20 and 335 normal training windows, whereas the models carry $2.0\text{--}5.5 \times 10^4$ parameters, so the comparison is made where deep models are structurally disadvantaged. Whether a crossover appears on genuinely long-running assets, where a deep model could see orders of magnitude more normal windows, is the central open question (Section 8).

7. DISCUSSION

7.1. What Changed the Story

Four corrections, each described where it is applied, change the conclusion of an earlier version of this study: the corrected failure label of the third IMS test (Appendix C), the onset-gated metric that defeats the latch-on exploit (Eq. 5), the $p < n$ channel-invariant feature space (Appendix B), and a significance test on the run rather than the within-run sampling factor (Section 4.9). Together they turn the earlier claim that SCADA averaging destroys lead time into a refutation on every dataset tested, and the previously reported significant positive effect into a non-significant trend.

7.2. Why the IMS Trend Appears but Is Neutral Elsewhere

The IMS trend has a clean candidate mechanism (Section 7.3): bin-averaging is a low-pass filter that removes single-window noise spikes, so the variance-sensitive control charts (3σ , EWMA) cross and hold their thresholds earlier than under decimation, which keeps the spikes—consistent with the largest IMS median shift appearing for 3σ , though not with a consistent direction across the three runs. An ordinary moving-average or Kalman filter on the decimated stream reproduces the trend (Appendix D.1), so the mechanism is generic low-pass smoothing rather than historian averaging specifically. ONGC is already smoothed at the 10 s historian rate, so further aggregation adds little, and the short, often abruptly failing XJTU bearings leave little quiet runway over which averaging could help. The synthesis is therefore not “averaging helps” but bounded non-destruction: across very different regimes nothing favours decimation beyond the operational margin, and where the signal is noisy and the runway long, aggregation tends to stabilize the health statistic.

The single-asset turbine case study. The one real historian record, an ONGC gas-turbine low-pressure-compressor bearing logged at a 10 s SCADA rate and ending in a genuine operator shutdown, is the cleanest illustration of the practical question because it needs no simulated coarsening: the aggregate-vs-decimate difference in raw lead is at most about a minute for every detector. Its detection quality is less clear-cut. Nine of its eleven detectors raise a raw alarm 30–35 h ahead of the labeled failure (RMS-trend 22.8 h, and Deep SVDD not before failure at all), but under the gated metric at $\tau = 10\%$ only Hotelling T^2 and RMS-trend are valid, because the others flag too many windows of a pre-onset region that ends only ~ 6 h before the shutdown; whether they are false alarms or genuine early detections that the late-biased onset cannot credit is undecidable at $n = 1$ (Appendix D.2). As a single asset it supports no inference and is excluded from the inferential family.

7.3. Why Aggregation Preserves What Decimation Loses

Bin-averaging by a factor f is a length- f moving-average filter followed by downsampling: an anti-alias filter, then decimation. Plain decimation is the same downsampling with the filter omitted. The prognostic content of the health statistic evolves on the degradation timescale—hours—far below the post-coarsening Nyquist rate even at $f = 20$, while measurement noise in the same series is broadband. Decimation therefore folds out-of-band noise into the low-frequency band occupied by the degradation trend, raising the variance of the health statistic without changing its mean; aggregation attenuates that noise before downsampling. For a detector that alarms when the statistic crosses $\mu_b + k\sigma_b$, a lower noise

variance means the threshold is crossed earlier and held more persistently.

The prediction that follows is about noise, not detectors: the advantage of aggregation should grow as the noise floor rises. It does, monotonically, from -0.35 h at the native noise level to $+6.60$ h at 10 dB, and an ordinary moving-average or Kalman filter on the decimated stream recovers it (Appendix D.1, Table 15).

The feature space degrades unevenly under aggregation, for a structural reason. Each channel-aggregated series is summarised over the rolling window by its mean, standard deviation, maximum and linear-trend slope. The mean and the slope are linear functionals of the underlying series, so block-averaging them commutes with the summary up to the weighting induced by the 50% window overlap; the standard deviation, the maximum and the cross-channel coherence are not linear, and block-averaging them is not equivalent to computing the summary on the coarsened series. Aggregation therefore preserves the amplitude and trend descriptors more faithfully than the dispersion and extremal ones.

We claim no conservation result. The detectors monitor a scalar collapse of the full standardised feature vector rather than any individual statistic, and the window overlap makes even the linear summaries a non-uniformly weighted average over each block. The argument states which descriptors degrade gracefully under aggregation and why the effect should track the noise floor; it does not predict the magnitude of the effect for any particular detector.

7.4. Practical Implications

For practitioners, storing bin-averaged summary statistics at SCADA rates, the default behavior of most historians, rather than keeping every f -th stored value changed bearing-fault warning time by less than the ± 1 h margin on every multi-bearing campaign tested (Section 6.4), and may improve it for control-chart detectors on noisy long-runway data. Conformal calibration adds a usable, distribution-free false-alarm guarantee wherever the normal region is genuinely stationary; the trade-off curve turns the gated metric into an operating-point selector for a stated budget; and the ablations caution that more features are not better under a false-alarm budget, time-domain amplitude and impulsiveness statistics carrying the usable signal on IMS. Where 3σ maximizes warning time within a 10% budget, operators whose cost structure penalizes false alarms heavily (for instance, process plants where each alarm triggers a mandatory manual inspection) have a complementary point in Deep SVDD, which stays within budget at $\leq 0.6\%$ pre-onset FAR across the swept thresholds on IMS (Table 9) with 14.8 h of warning and near-zero false alarms. Neither should be used on IMS test 1, which the conformal machinery flags as uncalibrated (Fig-

ure 4a).

8. LIMITATIONS AND THREATS TO VALIDITY

External validity (generalization). The IMS “aggregation helps” tendency is a non-significant positive-median trend, sign-consistent only for CUSUM and Hotelling T^2 ($n = 3$), that does not replicate on the larger multi-bearing sets, so we claim only non-destruction as the cross-dataset headline, not a positive effect; ONGC ($n = 1$) is a descriptive case study with no inference. The datasets, while deliberately diverse, are all rolling-element bearings; we do not claim the result transfers to gearboxes, blades or other failure modes without further testing.

A structural asymmetry limits what the cross-dataset null can establish. The three campaigns that supply statistical power—XJTU-SY, FEMTO/PRONOSTIA and University of Ferrara—comprise bearings whose total lifetimes span roughly 0.7 to 8.9 h, so an aggregation effect of the magnitude observed on IMS (a run-level median of $+15.1$ h for the 3σ chart) could not have been observed on them even if present. Their null therefore bounds the effect at the scale of minutes, which is what our equivalence margin reflects, not at the scale of the IMS result, while the only dataset whose runway is long enough to express such an effect has $n = 3$ and cannot reach significance at any margin. We therefore claim non-destruction at short bearing lifetimes, where the evidence is strong, and consistency with—not demonstration of—non-destruction at the multi-week lifetimes typical of industrial assets.

Residual gaming surfaces of the metric. We claim the metric defeats the specific latch-on (constant-on) exploit, not that it is unconditionally ungameable, and name two surfaces we do not close. (a) A detector could align its threshold to fire just after t_o^+ ; the onset uses only training-region statistics, so this cannot use test-region information, but an adversary who knows the onset rule could still tune toward it. (b) The onset health indicator shares its RMS/kurtosis basis with several detector features, so onset and detection are not fully independent, a circularity that could in principle favor detectors aligned to the onset statistic. Closing both (e.g. with an onset signal disjoint from the detector features) is future work.

Construct validity (does the metric measure warning time?). The onset, although leakage-free and stable to $k \in \{3, 4, 5\}$, is not the ground-truth defect-initiation time, which is unobservable in these datasets. A late onset, as on IMS test 1 where degradation and failure nearly coincide, inflates the pre-onset region, can mark a genuine early detection invalid and breaks the conformal exchangeability assumption (Figure 4a). Bearings whose physics provides essentially no warning runway score zero valid alarms regardless of detector (Table 7), which is correct behavior but makes the valid-alarm fraction partly a property of the bearings, not

only of the methods.

Internal validity (is the comparison fair?). One fixed hyperparameter protocol avoids optimistic per-condition tuning, the full trade-off sweep (Table 9) exposes threshold sensitivity rather than one hand-picked operating point, and the controlled sweep isolates information loss from window counting.

Statistical-conclusion validity. Confidence intervals are wide on IMS ($n = 3$); we resample runs, the unit of generalization, not seeds. An earlier pooled 15-pair Wilcoxon test treated the within-run factors as independent observations and thereby overstated significance; the run-level exact sign test cannot fall below $p = 0.25$ at $n = 3$, and after Holm correction across the forty-four-test family nothing is significant (Table 5), so the IMS result is a directional trend to be read with the multi-bearing nulls, never a significant effect. The test itself sets the minimum dataset: significance needs at least six sign-concordant runs ($2(\frac{1}{2})^6 = 0.031$), so the three IMS runs are at least a factor of two short—the quantitative content of the long-runway, many-bearing campaign that future work calls for. Such public datasets are rare; the closest single-bearing example is a KAIST bearing with a ~ 43 -hour lifecycle ($n = 1$, insufficient for inference). Groups with long-running industrial historian archives (gas turbines, compressors, paper machines) can apply the released pipeline directly; loader templates for every public dataset format used here are included in the repository at <https://github.com/Ansh-Goyal01/Scada-Leadtime-Benchmark> to minimize integration effort.

Limited external benchmarking. We do not benchmark the gated lead-time metric against the RUL-estimation literature. The comparison is not straightforward—RUL methods predict a remaining-life quantity and are scored by prediction error, whereas our metric scores when an alarm fires against a false-alarm budget—but the consequence is that the metric’s behaviour relative to that body of work is untested, and readers coming from RUL benchmarking should treat the two as complementary rather than ranked.

Future work. Two extensions would sharpen these conclusions most directly: run-to-failure campaigns on long-life bearings typical of industrial service, whose multi-week runway could express an aggregation effect that short laboratory bearings cannot; and comparable work on other bearing types and geometries, since the four inferential datasets are all rolling-element bearings under approximately steady speed and load and the single real-asset record is one gas-turbine compressor bearing. The complementary study coarsens the raw waveform itself, where averaging is a low-pass filter on the impulsive transients of an incipient fault (Section 4.8); our claim concerns historian summary statistics and does not cover it.

9. CONCLUSION

We presented a lead-time-centric evaluation protocol for bearing anomaly detection, whose onset-gated, false-alarm-budgeted metric defeats the latch-on exploit, and used it to test whether SCADA-rate logging destroys detection lead time. Across four inferential run-to-failure datasets (XJTU-SY $n = 10$, FEMTO/PRONOSTIA $n = 6$, University of Ferrara $n = 6$ and NASA IMS $n = 3$) and a real single-asset turbine SCADA case study (Appendix D.2), coarse aggregation does not destroy lead time; on IMS it shows a positive median shift for most control charts that is not consistent across the three runs, not significant at $n = 3$ and not surviving multiple-comparison correction across the $N = 44$ family. No dataset $\times$ detector cell shows decimation superior beyond a ± 1 h operational margin. That null is bounded by the scale of the evidence: the three campaigns supplying the statistical power comprise bearings of 0.7–8.9 h total life, on which an aggregation effect of the IMS magnitude could not have been observed, so the result bounds the effect at the scale of minutes and the multi-week runway remains untested. On the short-lived bearings tested, the deep reconstruction models do not overtake the control charts at any training fraction in $[0.20, 0.60]$; whether a crossover exists on long-runway assets is left open. The contribution is a reproducible methodology and a counter-intuitive empirical result: non-destruction, not a guaranteed positive effect. Future work will learn the degradation onset jointly with the detector, so that the metric’s anchor no longer depends on a hand-set health indicator (closing the circularity noted in Section 8), and extend the conformal guarantee to the non-exchangeable early-degradation regime in which it currently fails. We release the full harness, detectors, tests, and the exact tables and figures of this paper to support replication and extension.

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DATA AND CODE AVAILABILITY

All code (detectors, leakage-free onset, gated metric, sampling sweep, conformal wrapper, run-level statistical harness and the noise/denoiser mechanism test) is released with a pinned environment (requirements.lock.txt, Python 3.13) and a test suite asserting the metric’s resistance to the latch-on exploit, the leakage-free onset and the run-level

sign-test/Holm machinery. Every table and figure in this paper is generated directly from the result files released in the repository under `results/tables/`; no number is hand-edited. `results/tables/MANIFEST.md` maps each table and figure to the exact file that backs it, and `paper/verify_tables.py` re-derives the printed values from those files and fails if any disagrees—it runs as part of the test suite, so the claim is checked rather than asserted. Where a result exists in more than one arm (an earlier feature schema, or results computed before the aggregation bin width was corrected), the superseded arm has been moved to `results/tables/superseded/` with a note on why, so a replicator cannot select it by mistake. Repository: <https://github.com/Ansh-Goyal01/Scada-Leadtime-Benchmark>. A versioned snapshot is archived on Zenodo at the concept DOI <https://doi.org/10.5281/zenodo.20719076>. The IMS (NASA), XJTU-SY, FEMTO/PRONOSTIA and University of Ferrara run-to-failure datasets are publicly available from their original providers; the repository ships the loaders and the derived per-bearing feature caches for all four inferential datasets—IMS, XJTU-SY, FEMTO/PRONOSTIA and University of Ferrara—so the sampling sweep, ablations, calibration and trade-off analyses can be rerun without re-downloading the raw archives. The ONGC gas-turbine SCADA record is proprietary and covered by a non-disclosure agreement. Withheld, and not redistributable: the raw vibration waveforms, the channel-level measurements from which the features are built, and all asset identifiers. Released, so that the $n = 1$ case study can be repeated rather than merely inspected, are the derived files behind every ONGC number in this paper (all paths relative to `results/tables/`): the degradation health-indicator series in `ongc_health_indicator.csv`, a single baseline-standardized scalar per 10 s timestamp aggregated over all four channels, with the run-level split, onset and failure markers in `ongc_onset_markers.csv`—the published onset of 2023-11-13 03:44 is recomputable from these two files alone, with no proprietary input, and the repository’s test suite asserts exactly that; per-detector lead time and pre-onset false-alarm rate, together with the full sampling sweep (both modes $\times$ five factors $\times$ eleven detectors, 110 rows), in `n20_rerun_long_ONGC.csv`, with its aggregated and interval forms in `benchmark_ONGC_aggregate_n20.csv` and `benchmark_ONGC_leadtime_ci_n20.csv`; the per-detector ONGC median differences of Table 6 in `n20_ongc_minutes.csv`; the paired-test outputs in `benchmark_ONGC_paired_test_n20.csv`; the conformal calibration series behind Figure 4d in `calibration_ONGC.csv` and `calibration_ONGC_pooled.csv`; and the onset reruns under the two decoupled health indicators in `rg3_rerun_long_ONGC_rms_kurt.csv` and `rg3_rerun_long_ONGC_pcal.csv`, both of which were

computed before the aggregation bin width was corrected, retained as provenance for the indicator comparison; the invariance they establish compares onset indicators within one arm, so it does not depend on that correction, and the aggregate-vs-decimate contrast is taken from the corrected files named above. The health indicator is an aggregate from which the four per-channel measurements cannot be recovered, and being baseline-standardized it discloses no absolute vibration amplitude. No inferential claim in this paper depends on ONGC: it is a single asset, reported descriptively, excluded from every run-level significance test and from the 33-cell cross-dataset equivalence result, which is computed on the three multi-bearing campaigns.

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APPENDIX A: REPRODUCIBILITY SETTINGS

Table 12a lists the fixed hyperparameters, taken verbatim from the released config.py, and Table 11 the architectures and training protocol of the four deep models, with trainable parameter counts measured at the 49-dimensional invariant feature width; one protocol applies to every detector, dataset, run and sampling factor, with no per-condition tuning. Table 12b gives the wall-clock training and per-window inference cost on the largest IMS run ($X_{\text{train}} = 631$, $X_{\text{test}} = 506$ windows, single CPU core, median of five repeats). Training is a one-off offline step, and inference is at most ~ 0.5 ms per window (LSTM-AE), more than four orders of magnitude below the fastest historian polling in-

Table 11. Deep-model architectures and training protocol, from the released `config.py` and model definitions; no per-dataset tuning, and every model runs its full epoch budget. Parameters are counted at the 49-dimensional input (Appendix B). All four use the Adam optimiser at learning rate 10^{-3} , batch size 32 and seed 42, with no early stopping. Deep SVDD scores single windows.

	LSTM-AE	TCN-AE	Transformer-AD	Deep SVDD
Structure	1-layer LSTM encoder + linear to latent; 1-layer LSTM decoder + linear to output	4 temporal blocks, 2 dilated causal convolutions each, residual, 1×1 head	2 encoder layers, linear input projection and reconstruction head	2 linear layers, bias-free, ReLU between
Hidden width	64	32 channels	$d_{\text{model}} = 32$; FFN 64	32
Latent width	16	—	—	8
Kernel / dilations	—	$k = 3; 1, 2, 4, 8$	—	—
Attention heads	—	—	2	—
Sequence length	30 windows	30 windows	30 windows	n/a (per-window)
Epochs	50	40	40	40
Dropout	0.1 [†]	0.1	0.1	—
Loss	MSE reconstruction	MSE reconstruction	MSE reconstruction	mean squared distance to the fixed center
Trainable parameters	54,657	29,681	20,305	1,824

[†]PyTorch applies recurrent dropout only between stacked LSTM layers, so at one layer the configured 0.1 has no effect on LSTM-AE.

Table 12. (a) Fixed hyperparameter protocol, from `config.py`; onset tolerances are in samples of the health-indicator series, and the pre-onset FAR is measured over the full window $t < t_o$ (Eq. 5). (b) Compute cost on the largest IMS run (631 train / 506 test windows, 49 features, one core of an Intel Core i5-13420H, median of five repeats): training time and inference in μ s per window.

(a) Protocol		
Setting	Value	
Feature window / overlap	10 samples / 50%	
Feature schema	channel-invariant, 49-dim	
Train-fit feature cap	top 50 (stratified)	
Train / calib. / test	0.50 / 0.10 / 0.40	
Pre-onset FAR region	full window $t < t_o$	
Onset health indicator	rms_kurt	
Onset method / persistence	terminal excursion / 5	
Onset $k\sigma$ / gap tolerance	4σ / 10 samples	
Alarm threshold / persistence	97.5th pct / 3 windows	
FAR budget τ	0.10	
Isolation Forest	200 trees, contam. auto	
Hotelling T^2	PCA rank 10, $\alpha = 0.01$	
EWMA / CUSUM	$\lambda=0.2, k=3 / k=0.5, h=5$	
Deep models	seq-len 30, 40–50 epochs	
Transformer	$d_{\text{model}}=32$, 2 heads, 2 layers	
FEMTO channels	horiz. + vert. accel.	
Conformal calib. fraction	0.10	
Bootstrap resamples	runs, $B=2000$	
Random seed	42	
Environment	Py 3.13, requirements.lock.txt	

(b) Compute					
Detector	Train	Inf.	Detector	Train	Inf.
3σ	< 1 ms	0.2	EWMA	< 1 ms	1.0
CUSUM	< 1 ms	2.1	RMS-trend	< 1 ms	0.3
Hotelling T^2	20 ms	1.5	One-class SVM	11 ms	26
Iso. Forest	0.73 s	120	Deep SVDD	2.27 s	2.3
LSTM-AE	49.5 s	513	TCN-AE	46.2 s	379
Transf.-AD	41.6 s	300			

Table 13. The six per-snapshot base statistics from which the 49-dim invariant space is built. Each is computed per physical channel on the snapshot waveform x (mean μ , std σ), then channel-aggregated (mean, max) and window-summarised (mean, std, max, slope).

Statistic	Family	Formula
RMS	rms	$\sqrt{\frac{1}{n} \sum_i x_i^2}$
Kurtosis	kurt	$\frac{1}{n} \sum_i \left(\frac{x_i - \mu}{\sigma} \right)^4 - 3$
Skewness	skew	$\frac{1}{n} \sum_i \left(\frac{x_i - \mu}{\sigma} \right)^3$
Peak-to-peak	p2p	$\max_i x_i - \min_i x_i$
Crest factor	crest	$\max_i x_i / \text{RMS}$
Variance	var	$\frac{1}{n} \sum_i (x_i - \mu)^2$

APPENDIX B: THE 49-DIMENSIONAL INVARIANT FEATURE SPACE

Each snapshot’s raw waveform yields the six per-channel time-domain statistics of Table 13. Each family is made channel-count-invariant by the channel mean and the channel max, and each aggregated series is summarised over the rolling window (10 samples, 50% overlap) by its mean, standard deviation, maximum and least-squares trend slope, giving $6 \times 2 \times 4 = 48$ features; a cross-channel RMS-coherence feature (the mean pairwise Pearson correlation of the physical RMS channels within the window) brings the total to 49. A train-fit variance top- k selection ($k \leq 50$) keeps $p < n$, every statistic is fit on the training split only, and feature naming and the exact construction are in the released code.

APPENDIX C: IMS TEST-3 RELABELING JUSTIFICATION

An earlier label placed the failure of the third IMS test on 2004-04-08, leaving 21.5% of the recorded data after the nominal “failure”—impossible for a run-to-failure series. We relabel it to the last recorded sample (2004-04-18 02:42), at

terval in our study (10 s), so streaming condition monitoring at native SCADA rates is not compute-bound for any of the eleven detectors.

Table 14. IMS aggregate—decimate lead-time difference under the original (2004-04-08) and corrected (2004-04-18) third-test failure labels, all ten detectors. Run differences are (1st_test, 2nd_test, 3rd_test) in hours, collapsing the five within-run sampling factors to one mean difference per run (Section 4.9). $n_+/n_-/0$ counts positive, negative and exactly-zero runs; zero-difference runs are excluded from the sign test, so n_{eff} is the effective sample size under the original label. “Floor” is the smallest two-sided exact sign-test p attainable at that n_{eff} . Both arms use the 49-dimensional invariant feature schema, so the contrast isolates the label.

Detector	Corrected label			Original label			n_{eff}	p (floor)
	Run diffs (h)	Med	$n_+/n_-/0$	Run diffs (h)	Med	$n_+/n_-/0$		
3σ	+25.81, +15.10, -1.00	+15.10	2/1/0	+25.81, +15.10, 0.00	+15.10	2/0/1	2	0.50 (0.50)
CUSUM	+4.15, +4.67, +4.27	+4.27	3/0/0	+4.15, +4.67, 0.00	+4.15	2/0/1	2	0.50 (0.50)
EWMA	-3.89, +4.33, +2.10	+2.10	2/1/0	-3.89, +4.33, 0.00	0.00	1/1/1	2	1.00 (0.50)
Hotelling T^2	+1.00, +1.17, +81.74	+1.17	3/0/0	+1.00, +1.17, +35.30	+1.17	3/0/0	3	0.25 (0.25)
Iso. Forest	-0.67, +16.87, +3.43	+3.43	2/1/0	-0.67, +16.87, 0.00	0.00	1/1/1	2	1.00 (0.50)
Deep SVDD	+8.93, 0.00, 0.00	0.00	1/0/2	+8.93, 0.00, 0.00	0.00	1/0/2	1	1.00 (1.00)
RMS-trend	-0.17, 0.00, +2.37	0.00	1/1/1	-0.17, 0.00, 0.00	0.00	0/1/2	1	1.00 (1.00)
LSTM-AE	-1.39, -0.42, +56.12	-0.42	1/2/0	-1.39, -0.42, +41.33	-0.42	1/2/0	3	1.00 (0.25)
TCN-AE	-1.39, -0.42, +9.00	-0.42	1/2/0	-1.39, -0.42, +9.00	-0.42	1/2/0	3	1.00 (0.25)
Transformer-AD	-1.39, -0.42, +68.79	-0.42	1/2/0	-1.39, -0.42, +48.15	-0.42	1/2/0	3	1.00 (0.25)

which the mean-RMS health indicator has ramped from its flat baseline (~ 0.068) to ~ 0.228 before the final machine-off reading: within the flat normal-region baseline band at the old label, but well above the 4σ onset band at the corrected one. The relabeling moves the third test toward harder, not easier, detection: a run that was a guaranteed miss for most detectors becomes evaluable, and because the aggregate-vs-decimate contrast is computed within each run, the relabeling cannot artificially inflate it. It affects only the single hardest run and is not a tuning choice. We present it as an author relabeling based on the health-indicator trajectory, not independently verified ground truth: we know of no published benchmark that corrects this label, so we do not claim it as a community-established correction. It is the labeling we use throughout.

The relabeling made our own claim harder to support, not easier. Because a reader may ask whether the label was chosen for the result it produces, we re-ran the entire IMS sweep under the original 2004-04-08 label and report both arms side by side (Table 14). The decisive case is 3σ : under the original label the third test is a guaranteed miss, both lead times are zero, the run’s collapsed difference is exactly 0.00 h, and the zero-difference rule of Section 4.9 leaves 3σ at two positive and zero negative runs. An earlier version of this work claimed that in no run does aggregation shorten the lead time of the magnitude-monitoring detectors. That claim would have been *true* under the original label; it is false only under the corrected one, where the third test contributes -1.00 h, and we have withdrawn it. The relabeling moved the evidence against the paper’s own claim rather than in its favor, the opposite of what a result-motivated relabeling would do.

The relabeling is also narrower than one might expect. Under the original label the third test is a guaranteed miss for six of the ten detectors, not for all ten: 3σ , CUSUM, EWMA, Isolation Forest, Deep SVDD, and RMS-trend all score $L = 0$ in both modes. The remaining four—Hotelling T^2 , LSTM-AE,

TCN-AE, and Transformer-AD—alarm early enough to precede even the earlier failure time, so they retain a positive lead in both modes and their third run survives the zero-difference exclusion.

The effective sample size is detector-dependent, and neither label can reach significance. Because the zero-difference exclusion operates per detector, the original label does not simply reduce IMS to $n = 2$: it leaves $n_{\text{eff}} = 3$ for the four detectors that still warn on the third test, $n_{\text{eff}} = 2$ for 3σ , CUSUM, EWMA and Isolation Forest, and $n_{\text{eff}} = 1$ for Deep SVDD and RMS-trend. The exact two-sided sign test floors at $p = 0.25, 0.50$ and 1.00 at those sample sizes, so under the original label six of the ten detectors could not have reached $\alpha = 0.05$ at any effect size, and the four with $n_{\text{eff}} = 3$ floor at $p = 0.25$, already above α . The corrected label retains every run, but $n = 3$ still bounds the attainable p at 0.25 , so no IMS detector reaches significance under either label and the best attainable p is 0.25 in both arms (Hotelling T^2). The label does not change the direction: no median flips sign between the arms, 3σ holding at $+15.10$ h and Hotelling T^2 at $+1.17$ h, and the Holm verdict of Section 6 and the headline are untouched.

APPENDIX D: MECHANISM ANALYSIS AND ONGC CASE STUDY

D.1 Generic Smoothing vs. Historian Averaging

If bin-averaging helps on IMS because it is a low-pass filter suppressing single-window noise in the health statistic, letting the alarm threshold be crossed earlier and held more persistently than under decimation, two predictions follow. We test both on IMS at sampling factor 5, controlling window content so that only the smoothing differs. Here aggregation and decimation act on the per-snapshot series before windowing, whereas the IMS sweep coarsens the windowed feature vectors (Section 4.8), so this corroborates the mechanism on a related but not identical operation.

Table 15. IMS mechanism tests at sampling factor 5, the four magnitude SPC charts. (a) Noise injection: aggregate–decimate lead-time difference (h) collapsed to one mean per run (tests 1, 2, 3; $n = 3$), its mean, and the runs with a positive sign. (b) Denoisers applied to the decimated stream: mean lead (h) over the twelve chart-runs (four charts $\times$ three runs) and the valid alarms among them, against raw aggregation and raw decimation.

(a) Noise injection			
SNR	Per-run [t1, t2, t3]	Mean	Sign
None (native)	−7.94, +2.62, +4.28	−0.35	2/3
30 dB	−0.44, −2.59, +5.95	+0.97	1/3
20 dB	+10.60, +0.53, +0.12	+3.75	3/3
10 dB	+14.99, +0.53, +4.28	+6.60	3/3

(b) Denoisers		
Stream / denoiser	Mean lead	Valid
Aggregate (historian)	75.6	7/12
Decimate (raw)	76.0	8/12
Decimate + median	69.6	8/12
Decimate + moving avg	76.0	8/12
Decimate + Kalman	75.6	8/12
Decimate + wavelet	73.2	6/12

Prediction 1: the aggregate advantage should grow with noise. We add zero-mean Gaussian noise to the per-snapshot channel statistics (RMS and kurtosis), scaled per channel, at signal-to-noise ratios of 30, 20 and 10 dB and recompute the aggregate–decimate lead time, collapsing the four magnitude-monitoring SPC charts to one mean difference per run ($n = 3$ per level) to avoid the pseudoreplication corrected in the main analysis. The mean advantage rises monotonically with noise (Table 15a), from −0.35 h at the native noise level to +6.60 h at 10 dB, and at the two highest noise levels all three runs share the positive sign. The lone large negative, the −7.94 h native-noise entry, is IMS test 1, the abrupt near-coincident-onset run whose degradation begins only ~ 12 h before failure, which also violates the conformal guarantee (Figure 4a) and is unwarnable under every feature set. This is the qualitative signature of a low-pass mechanism, the noisier the signal the more bin-averaging helps, but at $n = 3$ it is a directional trend, not a significant effect (the exact sign test floors at $p = 0.25$, and at 30 dB the per-run signs are mixed): it corroborates the mechanism and cannot, at this sample size, establish it statistically.

Prediction 2: any smoother on the decimated stream should recover the advantage. We compare a median filter, a moving average, a constant-velocity Kalman filter and a wavelet denoiser, each applied to the decimated stream, against raw aggregation and raw decimation (Table 15b). The moving average and Kalman filter recover essentially the full aggregate lead time (76.0 and 75.6 h vs. aggregation’s 75.6 h), while the median and wavelet filters fall slightly short, so the effect is generic smoothing of the health signal, not a property unique to historian bin-averaging: an operator who decimates and

then lightly smooths recovers the same warning time. The wavelet denoiser drops to 6/12 valid alarms, below even raw decimation, because soft-thresholding the detail coefficients perturbs the pre-onset baseline enough to lift the pre-onset false-alarm rate past the budget: smoothing that preserves the normal baseline recovers the lead time, while smoothing that distorts it can cost validity. The one-alarm gap between raw decimation (8/12) and aggregation (7/12) at statistically indistinguishable lead times (76.0 vs. 75.6 h) is sampling noise at $n = 12$, not evidence that decimation is preferable.

D.2 Single-Asset ONGC Case Study

The ONGC record is a real Solar-Turbine low-pressure-compressor (LPC) dataset: four vibration channels logged at a 10 s SCADA rate over ~ 5 days (42,698 rows), ending in a genuine operator shutdown on 2023-11-13 09:46. This is authentic historian data with a real, documented event—the maintenance record confirms the shutdown followed a rolling-element bearing fault on the LPC—and the data-derived degradation onset (2023-11-13 03:44) precedes the operator’s action by roughly six hours. ONGC is the only dataset that requires no simulated coarsening to be SCADA-rate: it is the historian.

On raw lead, nine of the eleven detectors alarm 30–35 h before the labeled failure (seven within 34.9–35.3 h, CUSUM at 33.9 h and Hotelling T^2 at 30.4 h), the naive RMS-trend baseline gives 22.8 h, and Deep SVDD does not alarm before failure. Under the gated metric, however, only Hotelling T^2 (pre-onset FAR 6.5%) and RMS-trend (0.1%) have valid alarms at $\tau = 10\%$; the other eight alarming detectors exceed the budget, with pre-onset FAR from 11.7% (Isolation Forest) to 95.2% (CUSUM). The data-derived onset lies only ~ 6 h before the shutdown, so their first alarms fall 28–29 h before it and the windows they flag there count as pre-onset false alarms under the paper’s own definition. That verdict is ambiguous here. The onset estimator is systematically late under gradual degradation (Section 4.3), so some alarms that precede the estimated onset may be genuine early detection the estimator cannot credit; with $n = 1$ and no ground-truth defect-initiation time, the case study cannot distinguish the two. The aggregate-vs-decimate contrast is computed on raw lead and is unaffected: the difference is at most about a minute for every detector (Table 6), so on data that already is a real historian stream the choice of logging mechanism does not materially change warning time—the cleanest possible illustration of the practical question, and consistent with “averaging is not the enemy.” Because ONGC is a single asset ($n = 1$), we treat it as a descriptive case study and draw no statistical inference from it. Its conformal calibration (Figure 4d) tracks the diagonal closely, consistent with a mostly-stationary normal region.